

Regulation of hippocampal mossy fiber-CA3 synapse function by a Bcl11b/C1ql2/Nrxn3(25b+) pathway

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Artemis Koumoundourou, Märt Rannap, Elodie De Bruyckere, Sigrun Nestel, Carsten Reißner, Alexei V. Egorov, Pentao Liu, Markus Missler, Bernd Heimrich, Andreas Draguhn, Stefan Britsch 

Institute of Molecular and Cellular Anatomy, Ulm University, Ulm, Germany • Institute for Physiology and Pathophysiology, Heidelberg University, Heidelberg, Germany • Department of Neuroanatomy, Institute of Anatomy and Cell Biology, Faculty of Medicine, University of Freiburg, Freiburg, Germany • Institute of Anatomy and Molecular Neurobiology, University of Münster, Münster, Germany • School of Biomedical Sciences, Li Ka Shing Faculty of Medicine, The University of Hong Kong, Hong Kong. Centre for Translational Stem Cell Biology, 17W, Science Park, Hong Kong

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Abstract

The transcription factor Bcl11b has been linked to neurodevelopmental and neuropsychiatric disorders associated with synaptic dysfunction. Bcl11b is highly expressed in dentate gyrus granule neurons and is required for the structural and functional integrity of mossy fiber-CA3 synapses. The underlying molecular mechanisms, however, remained unclear. We show that the synaptic organizer molecule C1ql2 is a direct functional target of Bcl11b that regulates synaptic vesicle recruitment and long-term potentiation at mossy fiber-CA3 synapses *in vivo* and *in vitro*. Furthermore, we demonstrate C1ql2 to exert its functions through direct interaction with a specific splice variant of neurexin-3, Nrxn3(25b+). Interruption of C1ql2-Nrxn3(25b+) interaction by expression of a non-binding C1ql2 mutant or by deletion of Nrxn3 in the dentate gyrus granule neurons recapitulates major parts of the Bcl11b as well as C1ql2 mutant phenotype, and interferes with C1ql2 targeting to the synapse. Together, this study identifies a novel C1ql2-Nrxn3(25b+)-dependent signaling pathway through which Bcl11b controls mossy fiber-CA3 synapse function. Thus, our findings contribute to the mechanistic understanding of neurodevelopmental disorders accompanied by synaptic dysfunction.

eLife assessment

The authors identify a new role for C1ql2 at moss fiber synapses in the hippocampus and find that C1ql2, whose expression is controlled by Bcl11b, controls the recruitment of synaptic vesicles to active zones and is necessary for synaptic plasticity. The data implicating C1ql2 involvement, using numerous viral/genetic rescue approaches, are largely **convincing**, while the experimental evidence supporting the role of the specific Nrxn3 splice variant is less complete. These data are **valuable**, building on prior discoveries of how Bcl11b, a disease-relevant molecule, further contributes to our understanding of mossy-fiber synaptic development.

Introduction

Disruptions in synaptic structure and function have been identified as a major determinant in the manifestation of various neurodevelopmental and neuropsychiatric disorders, such as autism spectrum disorder, schizophrenia and intellectual disability (Hayashi-Takagi, 2017; Lepeta et al., 2016; Zoghbi & Bear, 2012). The need to understand how synaptic function is compromised in these disorders has accentuated the importance of studying the regulatory mechanisms of physiological synaptic function. These mechanisms involve cell adhesion molecules at both the pre- and post-synaptic side that act as synaptic organizers, whose unique combination determines the structural and functional properties of the synapse. Many such proteins have already been identified and our understanding of their complex role in synapse assembly and function has significantly increased over the last years (de Wit & Ghosh, 2016; O'Rourke et al., 2012; Sudhof, 2017). Furthermore, recent advances in the genetics of neurodevelopmental and neuropsychiatric disorders have implicated genes encoding for several of the known synaptic proteins, supporting a role for these molecules in the pathogenesis of corresponding disorders (Sudhof, 2021; Torres et al., 2017; Wang et al., 2018).

The sensitivity of the functional specification of the synapse to the combination of distinct synaptic proteins and their relative expression levels suggests that genetically encoded programs define at least facets of the synaptic properties in a cell-type specific manner (Sudhof, 2017, 2021). In fact, several of the synaptic proteins have been shown to promote formation of functional pre- and postsynaptic assemblies when presented in non-neuronal cells (Dalva et al., 2000; Dean et al., 2003; Scheiffele et al., 2000), showing that their ability to specify synapses is in part independent of signaling processes and neuronal activity and supporting the idea that synaptic function is governed by cues linked to cellular origin (Gomez et al., 2021). Thus, the investigation for synaptic organizers and their function in health and disease should be expanded to the transcriptional programs that regulate their expression.

Bcl11b (also known as Ctip2) is a zinc finger transcription factor that has been implicated in various disorders of the nervous system including Alzheimer's and Huntington's disease, and schizophrenia (Kunkle et al., 2016; Song et al., 2022; Whitton et al., 2018; Whitton et al., 2016). Patients with *BCL11B* mutations present with neurodevelopmental delay, overall learning deficits as well as impaired speech acquisition and autistic features (Eto et al., 2022; Lessel et al., 2018; Punwani et al., 2016; Yang et al., 2020). Bcl11b is expressed in several neuron types, including the dentate gyrus granule neurons (DGN) of the hippocampus. Expression of Bcl11b in the DGN starts during embryonic development and persists into adulthood (Simon et al., 2020). We have previously demonstrated that Bcl11b plays a crucial role in the development of the hippocampal mossy fiber system, the adult hippocampal neurogenesis as well as hippocampal learning and memory (Simon et al., 2016; Simon et al., 2012). In the mature hippocampus, Bcl11b is critical for the structural and functional integrity of mossy fiber synapses (MFS), the connection between DGN and CA3 pyramidal neurons (De Bruyckere et al., 2018). MFS have a critical role in learning and memory stemming from their unique structural and functional properties, such as an enormous pool of releasable synaptic vesicles (SV), and reliable presynaptic short- and long-term plasticity (Nicoll & Schmitz, 2005; Rollenhagen & Lubke, 2010). Conditional ablation of *Bcl11b* in murine DGN impairs presynaptic recruitment of SV and abolishes mossy fiber long-term potentiation (MF-LTP) (De Bruyckere et al., 2018). The molecular mechanisms, however, through which the transcriptional regulator Bcl11b controls highly dynamic properties of the MFS remained elusive.

In the present study, we show that the secreted synaptic organizer molecule C1ql2, a member of the C1q-like protein family (Yuzaki, 2017), is a functional target of Bcl11b in murine DGN. Reintroduction of C1ql2 in *Bcl11b* mutant DGN rescued the localization and docking of SV to the

active zone (AZ), as well as MF-LTP that was abolished upon *Bcl11b* ablation. Knock-down (KD) of *C1ql2* in wildtype animals recapitulated major part of the MFS phenotype observed in *Bcl11b* mutants. Furthermore, we show that *C1ql2* requires direct interaction with a specific neurexin-3 isoform, *Nrxn3(25b+)*, a member of a polymorphic family of presynaptic cell adhesion molecules (Reissner et al., 2013 [↗](#); Sudhof, 2017 [↗](#)), to recruit SV *in vitro* and *in vivo*. Finally, we demonstrate that the *C1ql2-Nrxn3(25b+)* interaction is necessary for the targeting of *C1ql2* to the MFS. Taken together, this study identifies a novel *Bcl11b/C1ql2/Nrxn3(25b+)*-dependent regulatory mechanism that is essential for the control of MFS function. Recent genetic studies suggested its single components to be associated with neurodevelopmental and neuropsychiatric disorders characterized by synaptic dysfunction. Our data, for the first time, demonstrate these molecules to be interconnected in one regulatory pathway. Thus, our findings provide new mechanistic insight into the pathogenesis of corresponding human disorders.

Results

Reintroduction of *C1ql2* into *Bcl11b* mutant dentate granule neurons restores synaptic vesicle recruitment at the mossy fiber-CA3 synapse

We demonstrated before that *Bcl11b* is critical for the structural and functional integrity of adult excitatory hippocampal MFS (De Bruyckere et al., 2018 [↗](#); Simon et al., 2012 [↗](#)). The downstream regulatory mechanisms, however, through which *Bcl11b* exerts its complex functions at the MFS remained unclear. Using differential transcriptomic analyses on *Bcl11b* conditional knock-out (cKO) and wildtype (WT) DGN we screened for candidate transcriptional targets of this transcription factor. Among these, we identified the synaptic organizer molecule *C1ql2*, previously implicated in modulating MFS functions (Matsuda et al., 2016 [↗](#)). *C1ql2* transcript and protein levels are massively downregulated in *Bcl11b* mutant DGN (**Supplementary figure 1a-b** [↗](#)), and the *Bcl11b* protein directly binds to consensus sequences within the *C1ql2* promotor (De Bruyckere et al., 2018 [↗](#)), suggesting *Bcl11b* to act on MFS through *C1ql2*. To directly test this, we stereotactically injected a *C1ql2*-expressing AAV (**Figure 1a-b** [↗](#)) into the dentate gyrus (DG) of *Bcl11b* cKO mice 2 weeks after induction of the mutation. AAV-mediated re-expression of *C1ql2* in the *Bcl11b* mutant DGN completely restored *C1ql2* protein expression to control levels (**Figure 1c-d** [↗](#); Control+EGFP: 1 ± 0.216 , *Bcl11b* cKO+EGFP: 0.2 ± 0.023 , *Bcl11b* cKO+EGFP-2A-*C1ql2*: 2.44 ± 0.745 , mean $\pm$ SEM). Furthermore, the spatial distribution of the exogenous *C1ql2* protein in mutants was indistinguishable from controls (**Figure 1e** [↗](#)). Using vGlut1 and Homer1 as pre- and postsynaptic markers, respectively, we observed exogenous *C1ql2* protein to precisely localize at glutamatergic synapses within the stratum lucidum (SL) of CA3, confirming that reintroduced *C1ql2* is correctly targeted to the MFS (**Figure 1f** [↗](#)).

MFS of *Bcl11b* cKO animals were characterized by a misdistribution of SV in relation to the AZ, with fewer SV being present in the vicinity of AZ (**Figure 2a** [↗](#)), as reflected by a lower average synapse score (**Figure 2b** [↗](#); c.f. De Bruyckere et al., 2018 [↗](#)). Reintroduction of *C1ql2* fully recovered the synapse score to control values (**Figure 2a-b** [↗](#); Control+EGFP: 3.4 ± 0.012 , *Bcl11b* cKO+EGFP: 2.96 ± 0.037 , *Bcl11b* cKO+EGFP-2A-*C1ql2*: 3.47 ± 0.043 , mean $\pm$ SEM). As revealed by the relative frequency of the individual synapse scores, *C1ql2* did not only reduce the number of inactive synapses, characterized by a synapse score of 0 but also improved the synapse score of active synapses (**Figure 2c** [↗](#)). To test for the specificity of the *C1ql2* effects, we overexpressed *C1ql3* in the DGN of *Bcl11b* cKO (**Figure 1a-b** [↗](#)). *C1ql3*, a different member of the *C1ql* subfamily, is co-expressed with *C1ql2* in DGN and the two proteins have been shown to form functional heteromers (Matsuda et al., 2016 [↗](#)). *C1ql3* expression is unchanged in *Bcl11b* cKO (**Supplementary figure 2a-b** [↗](#); Control+EGFP: 1 ± 0.022 , *Bcl11b* cKO+EGFP: 1.09 ± 0.126 , *Bcl11b* cKO+EGFP-2A-*C1ql2*: 0.87 ± 0.146 , mean $\pm$ SEM). Overexpression of *C1ql3* in the DGN of *Bcl11b* cKO

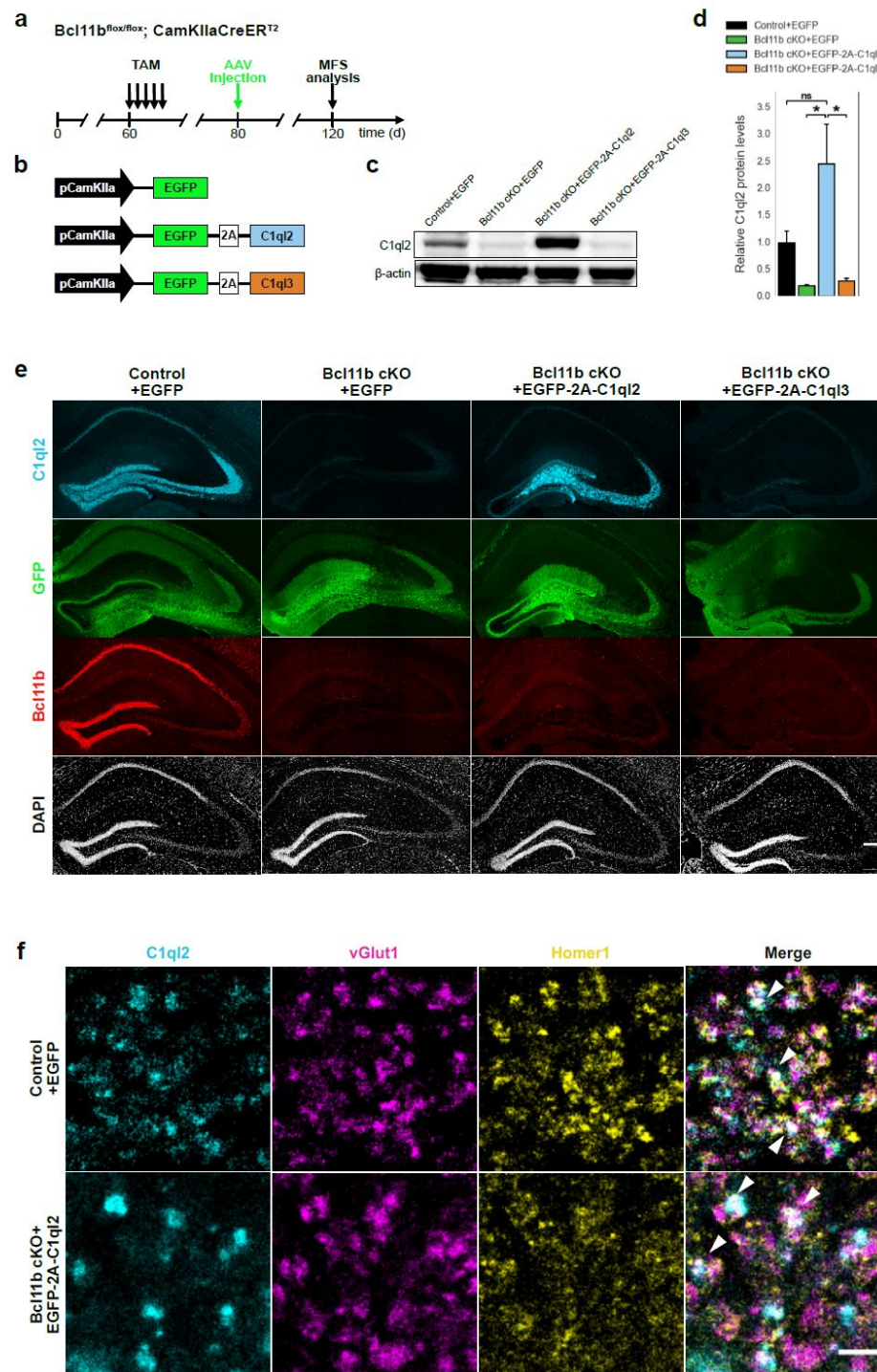


Figure 1.

Stereotaxic injection of C1ql2-expressing AAV into *Bcl11b* cKO DGN restores C1ql2 levels

a Experimental design to analyze the functions of C1ql2 in the MFS as a downstream target of Bcl11b. **b** AAV constructs injected in the DG of *Bcl11b* cKO and control littermates. **c** Western blot and **d** relative C1ql2 protein levels in mouse hippocampal homogenates. $n=3$. All data are presented as means; error bars indicate SEM. Two-way ANOVA and Tuckey's PHC. Bcl11b cKO+EGFP-2A-C1ql2 vs. Bcl11b cKO+EGFP: $*p=0.015$; Bcl11b cKO+EGFP-2A-C1ql2 vs. Bcl11b cKO+EGFP-2A-C1ql3: $*p=0.019$; ns, not significant. **e** Immunohistochemistry of C1ql2 (cyan), GFP (green) and Bcl11b (red) on hippocampal sections. Scale bar: 200 μ m. **f** Immunohistochemistry of C1ql2 (cyan), vGlut1 (magenta) and Homer1 (yellow) in the SL of CA3. White arrowheads indicate co-localizing puncta of all three proteins. Scale bar: 15 μ m

neither interfered with C1ql2 expression levels (**Figure 1c-e**; Bcl11b cKO+EGFP-2A-C1ql3: 0.29 ± 0.042 , mean $\pm$ SEM) nor was it able to rescue the synapse score of *Bcl11b* mutants (**Figure 2a-c**; Bcl11b cKO+EGFP-2A-C1ql3: 2.97 ± 0.062 , mean $\pm$ SEM).

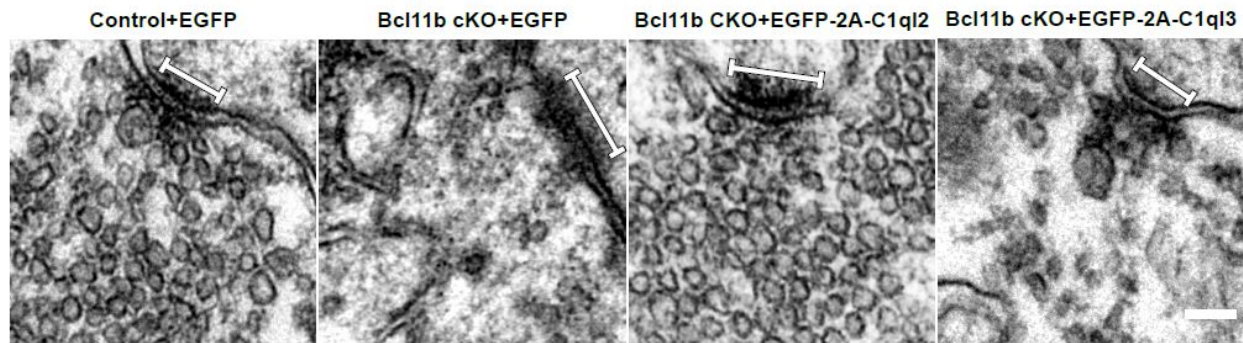
To analyze the C1ql2-dependent functions of Bcl11b on SV distribution in more detail, we quantified the number of SV docked on AZ in control animals, *Bcl11b* mutants, and upon the reintroduction of C1ql2. SV with a ≤ 5 nm distance from the plasma membrane were considered docked (Kusick et al., 2022; Vandael et al., 2020). *Bcl11b* mutant animals had significantly fewer docked vesicles per 100 nm of AZ profile length compared to control animals and more AZ with no docked vesicles at all. Rescue of C1ql2 expression restored the number of docked SV to control levels, while the overexpression of C1ql3 did not affect the number of docked vesicles (**Figure 2d, f-g**; Control+EGFP: 0.53 ± 0.098 , Bcl11b cKO+EGFP: 0.24 ± 0.038 , Bcl11b cKO+EGFP-2A-C1ql2: 0.51 ± 0.049 , Bcl11b cKO+EGFP-2A-C1ql3: 0.26 ± 0.041 , mean $\pm$ SEM). In contrast to vesicle distribution, Bcl11b and C1ql2 did not affect the size of the docked SV (**Figure 2e**; Control+EGFP: 36.39 ± 1.75 , Bcl11b cKO+EGFP: 35.18 ± 1.13 , Bcl11b cKO+EGFP-2A-C1ql2: 36.35 ± 1.01 , Bcl11b cKO+EGFP-2A-C1ql3: 36.65 ± 0.1 , mean $\pm$ SEM). Thus, our data suggest that C1ql2 specifically controls SV recruitment downstream of Bcl11b.

Conditional deletion of *Bcl11b* in the adult hippocampus also leads to a loss of MFS, as well as reduced ultrastructural complexity of the remaining mossy fiber boutons (MFB) (De Bruyckere et al., 2018), posing the question of whether these phenotypic features also depend on C1ql2. Interestingly, C1ql2 reintroduction in the DGN of *Bcl11b* cKO neither restored the loss of glutamatergic synapses, as quantified by the colocalization of pre- and postsynaptic markers, vGlut1 and Homer1 (**Supplementary figure 2e**; Control+EGFP: 90.65 ± 8.25 , Bcl11b cKO+EGFP: 60.68 ± 4.62 , Bcl11b cKO+EGFP-2A-C1ql2: 56.84 ± 6.99 , mean $\pm$ SEM), nor the reduced ultrastructural complexity of MFB, as quantified by the MFB perimeter/area ratio (**Supplementary figure 2c-d**; Control+EGFP: 0.0051 ± 0.00031 , Bcl11b cKO+EGFP: 0.0042 ± 0.00014 , Bcl11b cKO+EGFP-2A-C1ql2: 0.0037 ± 0.00021 , mean $\pm$ SEM). This suggests that Bcl11b acts on MFS through C1ql2-dependent as well as -independent signaling pathways.

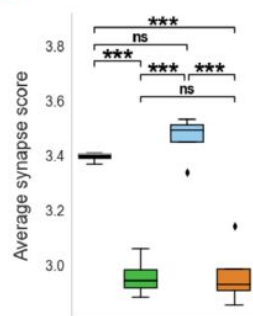
Reintroduction of C1ql2 into Bcl11b mutant dentate granule neurons rescues mossy fiber synapse long-term potentiation

The ultrastructural changes at the MFS point towards potential alterations in synaptic function. Indeed, adult-induced *Bcl11b* cKO was previously found to result in a loss of MF-LTP (De Bruyckere et al., 2018). We therefore tested whether the reintroduction of C1ql2 in *Bcl11b* cKO DGN can rescue LTP at the mutant MFS, similarly to SV recruitment. We stimulated mossy fibers in acute slices and measured the resulting field potentials in the SL of CA3. Field responses were carefully validated for the specificity of mossy fiber signals by the presence of strong paired-pulse facilitation and block by the mGluR antagonist DCG-IV. Under these conditions, input-output curves of fEPSP slopes versus axonal fiber volleys revealed no significant differences between control and *Bcl11b* cKO mice (**Supplementary figure 2f-g**; Control+EGFP: 1.95 ± 0.09 , Bcl11b cKO+EGFP: 2.02 ± 0.14 , mean $\pm$ SEM), indicating that basal synaptic transmission was unaltered in the *Bcl11b* mutants. We then induced LTP by high-frequency stimulation (HFS) of mossy fibers in control and *Bcl11b* cKO animals with or without AAV-mediated expression of C1ql2. Compared to controls, *Bcl11b* mutants injected with the control AAV displayed a strong reduction of LTP at 20-30 and 30-40 min after induction (**Figure 3a-c**; 0-10 min: Control+EGFP: 90.4 ± 7.2 , Bcl11b cKO+EGFP: 106.1 ± 10.8 , 10-20 min: Control+EGFP: 42.7 ± 3.6 , Bcl11b cKO+EGFP: 39.5 ± 4.6 , 20-30 min: Control+EGFP: 52.5 ± 7.6 , Bcl11b cKO+EGFP: 24.8 ± 3.2 , 30-40 min: Control+EGFP: 50.1 ± 7.3 , Bcl11b cKO+EGFP: 20.3 ± 3.7 , mean $\pm$ SEM), consistent with our previous data (De Bruyckere et al., 2018). The loss of LTP was completely reversed upon the re-expression of C1ql2, with the rescue mice exhibiting comparable LTP to controls at all time intervals (**Figure 3a-c**; 0-10 min: Control+EGFP: 90.4 ± 7.2 , Bcl11b cKO+EGFP-2A-C1ql2: 86.5 ± 7.4 , 10-20 min: Control+EGFP: 42.7 ± 3.6 , Bcl11b cKO+EGFP-2A-C1ql2: 49.4 ± 5.9 , 20-30 min: Control+EGFP: 52.5 ± 7.6 , Bcl11b cKO+EGFP-2A-

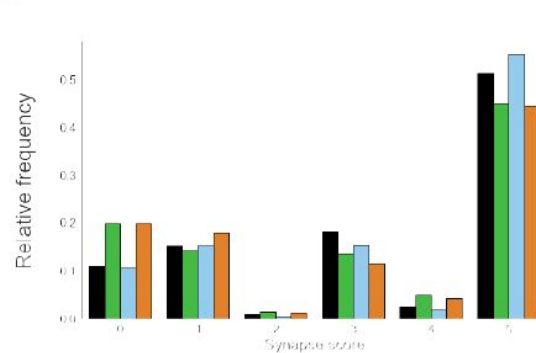
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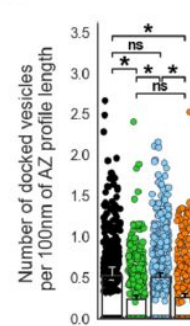
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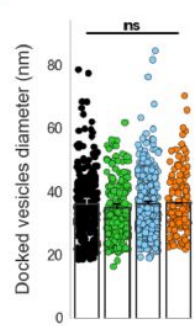
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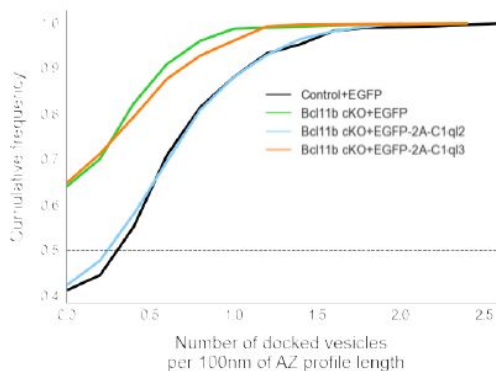


e



■ Control+EGFP ■ Bcl11b cKO+EGFP ■ Bcl11b cKO+EGFP-2A-C1qI2 ■ Bcl11b cKO+EGFP-2A-C1qI3

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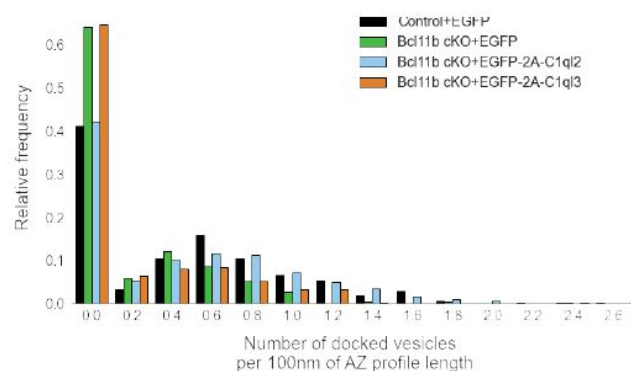


Figure 2.

C1qI2 reintroduction in *Bcl11b* cKO DGN rescues SV recruitment in MFS

a Electron microscope images of MFS and proximal SV. White bars mark synapse length from the postsynaptic side. Scale bar: 100 nm. **b** Average synapse score. Control+EGFP, n=3; Bcl11b cKO+EGFP, Bcl11b cKO+EGFP-2A-C1qI2, Bcl11b cKO+EGFP-2A-C1qI3, n=4. Two-way ANOVA and Tukey's PHC. Control+EGFP vs. Bcl11b cKO+EGFP: *** $p=0.0002$, and vs. Bcl11b cKO+EGFP-2A-C1qI3: *** $p=0.0003$; Bcl11b cKO+EGFP-2A-C1qI2 vs. Bcl11b cKO+EGFP and vs. Bcl11b cKO+EGFP-2A-C1qI3: *** $p<0.0001$; ns, not significant. **c** Relative frequency of synapse scores. **d** Number of docked vesicles per 100 nm AZ profile length. Control+EGFP, Bcl11b cKO+EGFP-2A-C1qI3, n=3; Bcl11b cKO+EGFP, Bcl11b cKO+EGFP-2A-C1qI2, n=4. All data are presented as means; error bars indicate SEM. Points represent the individual examined AZ and SV, respectively. Two-way ANOVA and Tukey's PHC. Control+EGFP vs. Bcl11b cKO+EGFP: * $p=0.024$, and vs. Bcl11b cKO+EGFP-2A-C1qI3: * $p=0.045$; Bcl11b cKO+EGFP-2A-C1qI2 vs. Bcl11b cKO+EGFP: * $p=0.026$, and vs. Bcl11b cKO+EGFP-2A-C1qI3: * $p=0.049$; ns, not significant. **e** Diameter of docked vesicles. Control+EGFP, Bcl11b cKO+EGFP-2A-C1qI3, n=3; Bcl11b cKO+EGFP, Bcl11b cKO+EGFP-2A-C1qI2, n=4; Two-way ANOVA. ns, not significant. **f** Cumulative and **g** relative frequency of the number of docked vesicles per 100 nm AZ profile length.

C1ql2: 47.2 ± 5.7 , 30-40 min: Control+EGFP: 50.1 ± 7.3 , Bcl11b cKO+EGFP-2A-C1ql2: 44.9 ± 5.3 , mean $\pm$ SEM). Importantly, this rescue effect was specific to C1ql2 as the overexpression of C1ql3 failed to reverse the Bcl11b cKO phenotype (**Figure 3a-c**; 0-10 min: Control+EGFP: 90.4 ± 7.2 , Bcl11b cKO+EGFP-2A-C1ql3: 104.2 ± 9.9 , 10-20 min: Control+EGFP: 42.7 ± 3.6 , Bcl11b cKO+EGFP-2A-C1ql3: 44.4 ± 5.7 , 20-30 min: Control+EGFP: 52.5 ± 7.6 , Bcl11b cKO+EGFP-2A-C1ql3: 29.0 ± 2.3 , 30-40 min: Control+EGFP: 50.1 ± 7.3 , Bcl11b cKO+EGFP-2A-C1ql3: 22.6 ± 2.4 , mean $\pm$ SEM).

MF-LTP is known to be mediated by a cAMP-dependent signaling cascade. To test whether Bcl11b acts on LTP through the cAMP-dependent pathway, we directly activated the cAMP pathway in slices from control and *Bcl11b* cKO mice by applying the adenylyl cyclase (AC) activator forskolin (Weisskopf et al., 1994). Compared to slices from control animals, forskolin-induced LTP in the mutants had a significantly lower peak and remained significantly weaker throughout the recording (**Figure 3d-f**; 15-30 min: Control: 342.4 ± 28.0 , Bcl11b cKO: 232.3 ± 17.7 , 105-120 min: Control: 88.0 ± 7.4 , Bcl11b cKO: 47.2 ± 8.2 , mean $\pm$ SEM). This suggests that the regulation of MF-LTP by Bcl11b involves the cAMP-dependent signaling pathway.

Knock-down of C1ql2 in dentate granule neurons perturbs synaptic vesicle recruitment and long-term potentiation at the mossy fiber-CA3 synapse

To further corroborate the observation that Bcl11b acts on MFS specifically through C1ql2, we knocked down *C1ql2* expression in DGN of adult WT mice by stereotactically injecting an AAV carrying an shRNA cassette against C1ql2 (**Figure 4a**). Quantitative PCR (**Figure 4b**), western blot analysis (**Figure 4c**) as well as immunohistochemistry using C1ql2 antibodies on hippocampal tissue (**Figure 4d**) revealed that the shRNA-mediated KD resulted in a strong reduction of the *C1ql2* transcript (**Figure 4b**; +shNS-EGFP: 1 ± 0.07 , +shC1ql2-EGFP: 0.23 ± 0.059 , mean $\pm$ SEM) as well as protein levels (**Figure 4c-d**), as compared to animals injected with the control AAV. The shRNA-mediated KD of *C1ql2* had no effect on the expression of C1ql3, demonstrating the specificity of this approach (**Supplementary figure 3a-b**; +shNS-EGFP: 1 ± 0.09 , +shC1ql2-EGFP: 0.986 ± 0.035 , mean $\pm$ SEM). Compared to controls, *C1ql2* KD was sufficient to reduce the average synapse score to similar levels as observed in *Bcl11b* cKO (**Figure 4e**, **Supplementary figure 3c**; +shNS-EGFP: 3.38 ± 0.069 , +shC1ql2-EGFP: 3.15 ± 0.031 , mean $\pm$ SEM), as well as the number of docked vesicles per 100 nm of AZ profile length (**Figure 4f**, **Supplementary figure 3d-e**; +shNS-EGFP: 0.48 ± 0.04 , +shC1ql2-EGFP: 0.31 ± 0.02 , mean $\pm$ SEM). At the same time, the diameter of the docked vesicles remained unchanged (**Supplementary figure 3f**; +shNS-EGFP: 34.28 ± 0.84 , +shC1ql2-EGFP: 35.37 ± 0.21 , mean $\pm$ SEM). To test whether shRNA-mediated KD of *C1ql2* expression also affects MF-LTP, we performed LTP recordings in *C1ql2* KD and control mice. Compared to controls, slices from *C1ql2* KD mice exhibited a significant reduction of LTP at 20-30 and 30-40 min time intervals, similarly to *Bcl11b* cKO animals (**Figure 4g-i**; 0-10 min: +shNS-EGFP: 105.0 ± 4.0 , +shC1ql2-EGFP: 94.3 ± 4.5 , 10-20 min: +shNS-EGFP: 56.3 ± 4.5 , +shC1ql2-EGFP: 35.1 ± 2.8 , 20-30 min: +shNS-EGFP: 50.2 ± 4.5 , +shC1ql2-EGFP: 23.4 ± 3.5 , 30-40 min: +shNS-EGFP: 44.6 ± 4.3 , +shC1ql2-EGFP: 20.1 ± 4.1 , mean $\pm$ SEM). Thus, KD of *C1ql2* in WT DGN recapitulates major phenotypes observed upon *Bcl11b* cKO, supporting that Bcl11b controls SV recruitment and LTP in hippocampal MFS specifically through C1ql2.

C1ql2-Nrxn3(25b+) interaction recruits presynaptic vesicles *in vitro* and *in vivo*

C1ql2 was previously shown to interact with a particular splice variant of Nrxn3 β containing exon 25b sequences, Nrxn3(25b+), which was recombinantly expressed in HEK293 cells (Matsuda et al., 2016). To explore whether the Bcl11b/C1ql2-dependent regulation of MFS involves interaction with neuronal Nrxn3, we co-cultured myc-C1ql2 expressing HEK293 cells with primary hippocampal neurons transfected with GFP-Nrxn3 α (25b+). We used here the extracellularly longer

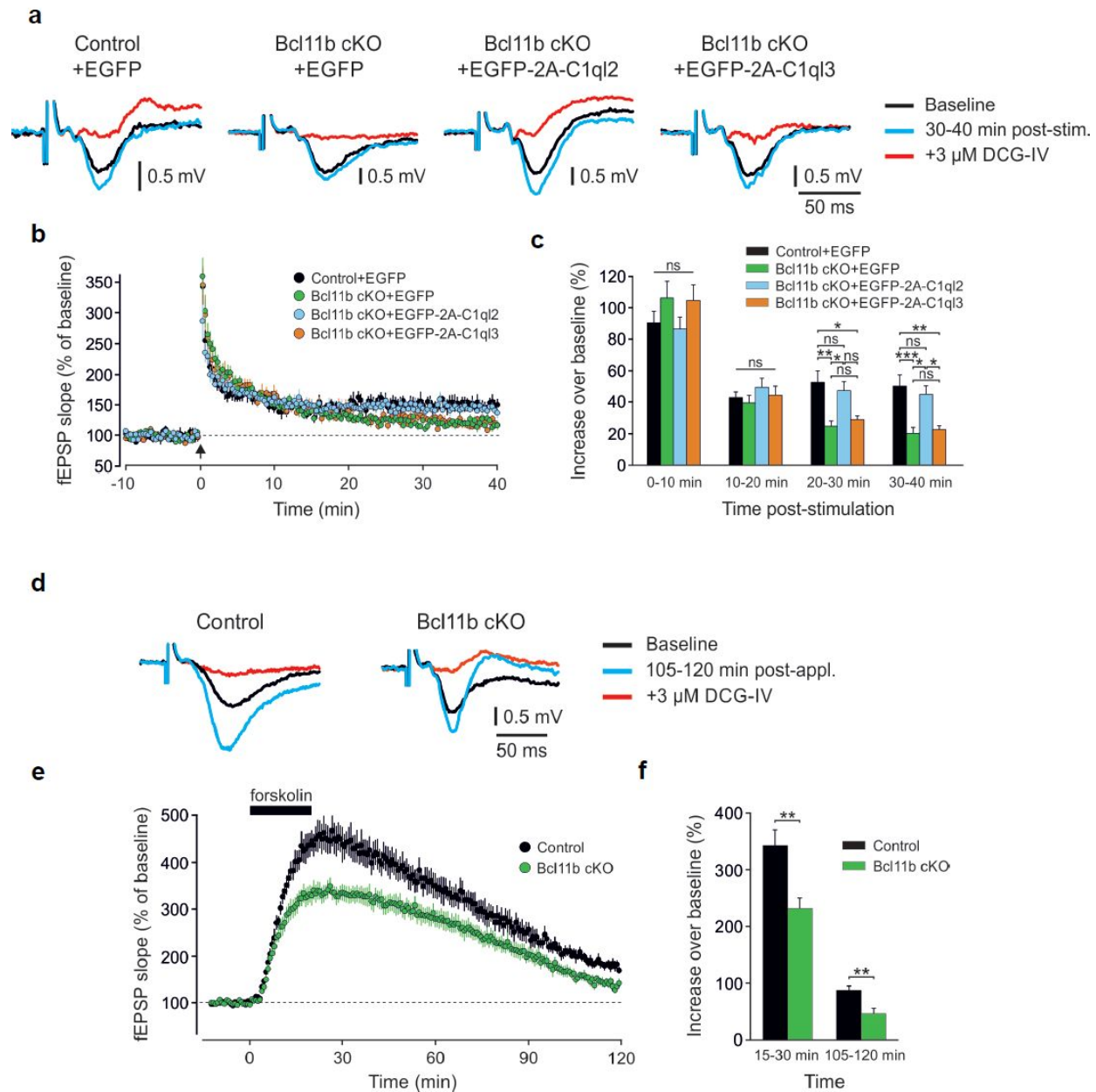


Figure 3.

C1ql2 reintroduction in *Bcl11b* cKO DGN rescues mossy fiber LTP

a Representative fEPSP traces showing baselines before HFS (black), fEPSP changes 30-40 min after HFS (cyan) and following the application of 3 μ M DCG-IV (red). **b** Time course of fEPSP slopes. The black arrow indicates HFS and the dashed line the baseline level. **c** Quantification of fEPSP facilitation at four different time intervals after HFS. Changes in fEPSP slope are shown as percentage of the mean baseline fEPSP. Control+EGFP, $n=7$; Bcl11b cKO+EGFP, $n=8$; Bcl11b cKO+EGFP-2A-C1ql3, $n=8$; Bcl11b cKO+EGFP-2A-C1ql2, $n=6$. All data are presented as means; error bars indicate SEM. One-way ANOVA followed by Bonferroni's PHC for each time interval. 20-30 min: Control+EGFP vs. Bcl11b cKO+EGFP: $^{**}p=0.002$, and vs. Bcl11b cKO+EGFP-2A-C1ql3: $^{*}p=0.011$; Bcl11b cKO+EGFP-2A-C1ql2 vs. Bcl11b cKO+EGFP: $^{*}p=0.023$; 30-40 min: Control+EGFP vs. Bcl11b cKO+EGFP: $^{***}p<0.001$, and vs. Bcl11b cKO+EGFP-2A-C1ql3: $^{**}p=0.002$; Bcl11b cKO+EGFP-2A-C1ql2 vs. Bcl11b cKO+EGFP: $^{*}p=0.01$ and vs. Bcl11b cKO+EGFP-2A-C1ql3: $^{*}p=0.0523$; ns, not significant. **d** Representative fEPSP traces showing baselines before forskolin application (black), fEPSP changes 105-120 min after the start of application (cyan) and following the addition of 3 μ M DCG-IV (red). **e** Time course of fEPSP slopes. The black solid line indicates forskolin perfusion and dashed line the baseline level. **f** Quantification of fEPSP facilitation at two different time intervals after the start of forskolin application. Changes in fEPSP slope are shown as percentage of the mean baseline fEPSP. Control, $n=8$. All data are presented as means; error bars indicate SEM. Unpaired t-test for both time intervals. 15-30 min: $^{**}p=0.005$; 105-120 min: $^{**}p=0.0025$.

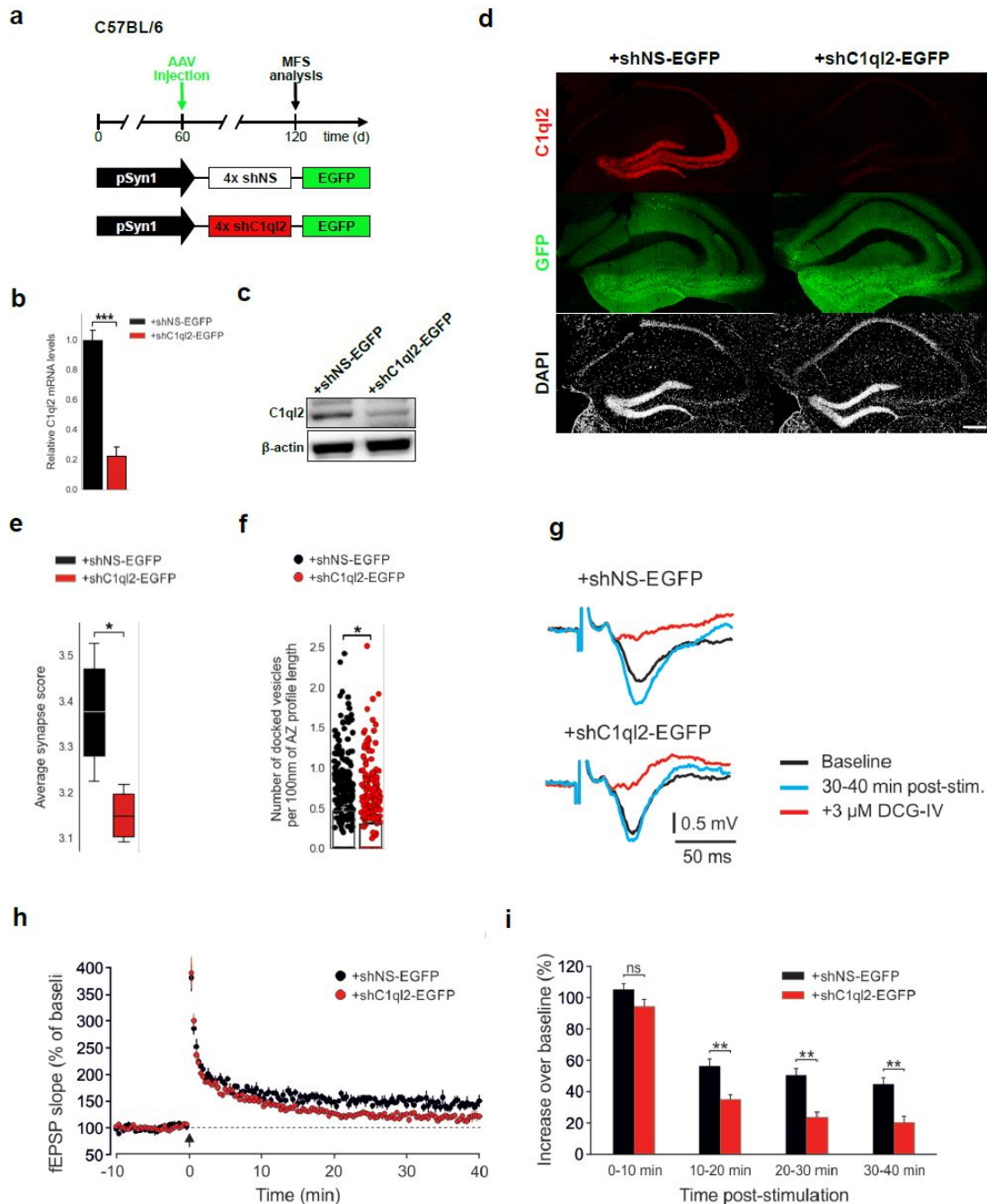


Figure 4.

KD of *C1ql2* in DGN of WT mice impairs SV recruitment and LTP

a Experimental design to analyze the MFS after AAV-mediated KD of *C1ql2* in WT DGN. **b** Relative *C1ql2* mRNA levels in DGN. $n=4$ (3). All data are presented as means; error bars indicate SEM. Unpaired t-test: *** $p=0.0002$. **c** Western blot of mouse hippocampal homogenates. **d** Immunohistochemistry of C1ql2 (red) and GFP (green) on hippocampal sections. Scale bar: 200 μm. **e** Average synapse score. $n=4$. Unpaired t-test. * $p=0.025$. **f** Number of docked vesicles per 100 nm AZ profile length. $n=3$. All data are presented as means; error bars indicate SEM. Points represent the individual examined AZ. Unpaired t-test. * $p=0.018$. **g** Representative fEPSP traces showing baselines before HFS (black), fEPSP changes 30-40 min after HFS (cyan) and following the application of 3 μM DCG-IV (red). **h** Time course of fEPSP slopes. The black arrow indicates HFS and the dashed line the baseline level. **i** Quantification of fEPSP facilitation at four different time intervals after HFS. Changes in fEPSP slope are shown as percentage of the mean baseline fEPSP. +shNS-EGFP, $n=6$; +shC1ql2-EGFP, $n=7$. All data are presented as means; error bars indicate SEM. Mann-Whitney U-test for each time interval. 10-20 min: ** $p=0.0012$; 20-30 min: ** $p=0.0023$; 30-40 min: ** $p=0.0023$; ns, not significant.

Nrxn3 α isoform because it is more strongly expressed in the murine DG compared to Nrxn3 β (Uchigashima et al., 2019). C1ql2-dependent recruitment of Nrxn3 α (25b+) was quantified by the surface area of HEK293 cells covered by GFP-Nrxn3 α (25b+)-positive neuronal profiles (**Figure 5a**). HEK293 cells expressing myc-C1ql2 had a significantly larger surface area covered by neuronal Nrxn3 α (25b+) in comparison to HEK293 cells expressing myc-tag only (**Figure 5b**; myc-C1ql2: 39.97 $\pm$ 3.99, myc-tag: 17.29 $\pm$ 2.27, mean $\pm$ SEM). Using vGlut1 immunoreactivity as a proxy for SV localization (Aoto et al., 2007; Freneau et al., 2004), we examined in the same system whether C1ql2-expressing HEK293 cells were able to cluster vGlut1 in surrounding GFP-Nrxn3 α (25b+)-positive neurons (**Figure 5c**). Interestingly, vGlut1 accumulation was significantly increased in GFP-Nrxn3 α (25b+)-positive neurons contacting myc-C1ql2 expressing HEK293 cells as compared to C1ql2-negative HEK293 cells (**Figure 5d**; myc-C1ql2: 40.88 $\pm$ 3.25, myc-tag: 24.78 $\pm$ 4.99, mean $\pm$ SEM). To specifically analyze whether endogenous Nrxn3 is required for C1ql2-mediated vGlut1 accumulation in neurons, we co-cultured myc-C1ql2 expressing HEK293 cells with primary hippocampal neurons derived from *Nrxn123^{flox/flox}* mice, in which all three *Nrxn* genes are floxed (Chen et al., 2017), and which we transfected with either active Cre recombinase or an inactive Cre (Klatt et al., 2021) (**Figure 5e**). Neurons lacking all Nrxn variants showed a significantly lower accumulation of endogenous vGlut1 when contacting the C1ql2-expressing HEK293 cells compared to control neurons. Strikingly, selective reintroduction of the Nrxn3 α (25b+) isoform into the null-mutant *Nrxn* cKO neurons was sufficient to normalize vGlut1 accumulation *in vitro* (**Figure 5f**; inactive Cre: 51.66 $\pm$ 5.97, Cre: 27.83 $\pm$ 2.83, Cre+ Nrxn3 α (25b+): 39.23 $\pm$ 4.3, mean $\pm$ SEM). Collectively, our data strongly suggest that the C1ql2-mediated recruitment of vGlut1-positive SV in hippocampal neurons depends on the presence of Nrxn3 α (25b+).

To explore the relevant epitope that mediates binding of C1ql2 to Nrxn3(25b+) proteins, we analyzed the solvent accessible electrostatic surface properties of the C1ql-domain trimeric structure of C1ql2 (Ressl et al., 2015) (PDB_ID: 4QPY) and found that a change of lysine262 (K262) to glutamic acid renders a large area underneath the C1ql2-specific calcium and receptor binding loops negative (**Figure 5g**) and hypothesized that this would repel binding to Nrxn3(25b+). We generated a C1ql2.K262E variant, expressed it in HEK293 cells as before, and tested it for its ability to cluster Nrxn3 α (25b+) as well as vGlut1 in contacting primary neurons (**Figure 5a-d**). In the presence of C1ql2.K262E, recruitment of Nrxn3 α (25b+) was significantly lower compared to WT C1ql2 and indistinguishable from myc-tag control levels. (**Figure 5a-b**; myc-K262E: 18.84 $\pm$ 5.15). Moreover, the expression of C1ql2.K262E in HEK293 cells was unable to accumulate vGlut1 in contacting neurons expressing GFP-Nrxn3 α (25b+) (**Figure 5c-d**; myc-K262E: 16.9 $\pm$ 1.2, mean $\pm$ SEM). Together, these results provide *in vitro* evidence that the clustering of vGlut1 depends on an intact C1ql2-Nrxn3(25b+) interaction, and that a single point mutation that creates a negative charge of that surface area underneath the C1ql2-specific calcium and receptor binding loops abolishes this binding activity and, thereby, the regulation of SV clustering.

To validate our identification of K262 as a key residue for the C1ql2-Nrxn3(25b+) interaction *in vivo*, we expressed C1ql2.K262E in *Bcl11b* cKO DGN, in which endogenous C1ql2 expression is downregulated by the ablation of *Bcl11b* (**Figure 6a**). AAV-mediated introduction of C1ql2.K262E in *Bcl11b* cKO DGN resulted in strong over all expression of the mutant protein (**Figure 6b-c**; Control+EGFP: 1 $\pm$ 0.42, Bcl11b cKO+EGFP-2A-K262E: 9.68 $\pm$ 4.75, mean $\pm$ SEM). However, the spatial distribution of C1ql2.K262E was notably different from the WT protein in the SL of CA3 where most of the MFS are located (**Figure 6d**). In the SL, expression levels of C1ql2.K262E were significantly lower compared to WT C1ql2 as quantified by the integrated fluorescence density (**Figure 6d-e**; Bcl11b cKO+EGFP-2A-C1ql2: 9.75 $\pm$ 0.57 $\times 10^4$, Bcl11b cKO+EGFP-2A-K262E: 5.89 $\pm$ 0.55 $\times 10^4$, mean $\pm$ SEM). As C1ql3 has been shown to form functional heteromers with C1ql2 at the MFS, we examined the spatial distribution of the C1ql3 protein upon AAV-mediated introduction of C1ql2.K262E for potential expression pattern changes but observed no difference (**Supplementary figure 4a**).

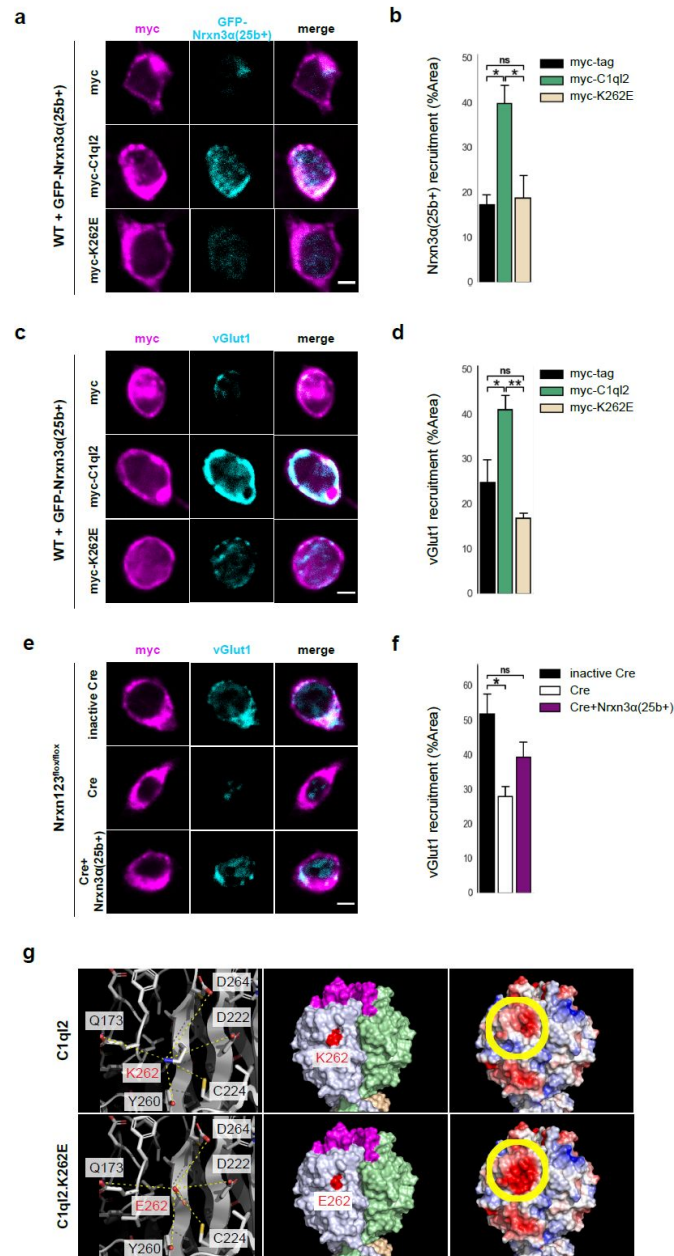


Figure 5.

C1ql2-Nrxn3 interaction recruits vGlut1 *in vitro*

a Immunocytochemistry of myc-tagged C1ql2, C1ql2.K262E or myc-tag (magenta) expressing HEK293 cells and GFP-Nrxn3a(25b+) (cyan) from contacting hippocampal neurons. Scale bar: 5 μ m. **b** Nrxn3a(25b+) recruitment by differentially transfected HEK293 cells. n=3. All data are presented as means; error bars indicate SEM. One-way ANOVA and Tuckey's PHC. myc-C1ql2 vs. myc-tag: *p=0.016, and vs. myc-K262E: *p=0.022; ns, not significant. **c** Immunocytochemistry of myc-tagged C1ql2, C1ql2.K262E or myc-tag (magenta) expressing HEK293 cells and vGlut1 (cyan) from contacting hippocampal neurons. Scale bar: 5 μ m. **d** vGlut1 recruitment by differentially transfected HEK293 cells. n = 3. All data are presented as means; error bars indicate SEM. One-way ANOVA and Tuckey's PHC. myc-C1ql2 vs. myc-tag: *p=0.04, and vs. myc-K262E: **p=0.007; ns: non-significant. **e** Immunocytochemistry of myc-tagged C1ql2 (magenta) expressing HEK293 cells and vGlut1 (cyan) from contacting control, *Nrxn123* KO or *Nrxn123* KO with *Nrxn3a*(25+) rescued hippocampal neurons. Scale bar: 5 μ m. **f** vGlut1 recruitment by HEK293 cells in presence or absence of neuronal Nrxns. n = 3. All data are presented as means; error bars indicate SEM. One-way ANOVA and Tuckey's PHC. inactive Cre vs. Cre: *p=0.023, and vs. Cre+ *Nrxn3a*(25+): p=0.21; ns: non-significant. **g** Trimeric structures of C1ql2 (PDB_ID: 4QPY, upper panels) and the variant C1ql2.K262E (lower panels). Residue 262 is the central residue (red, left & middle panels) of a larger area underneath the C1ql2-specific calcium and receptor binding loops (magenta, middle panel). The mutation K262E alters the charge of that surface area negative (yellow-circled area, right panels) and makes it potentially repulsive to bind *Nrxn3*(25b+).

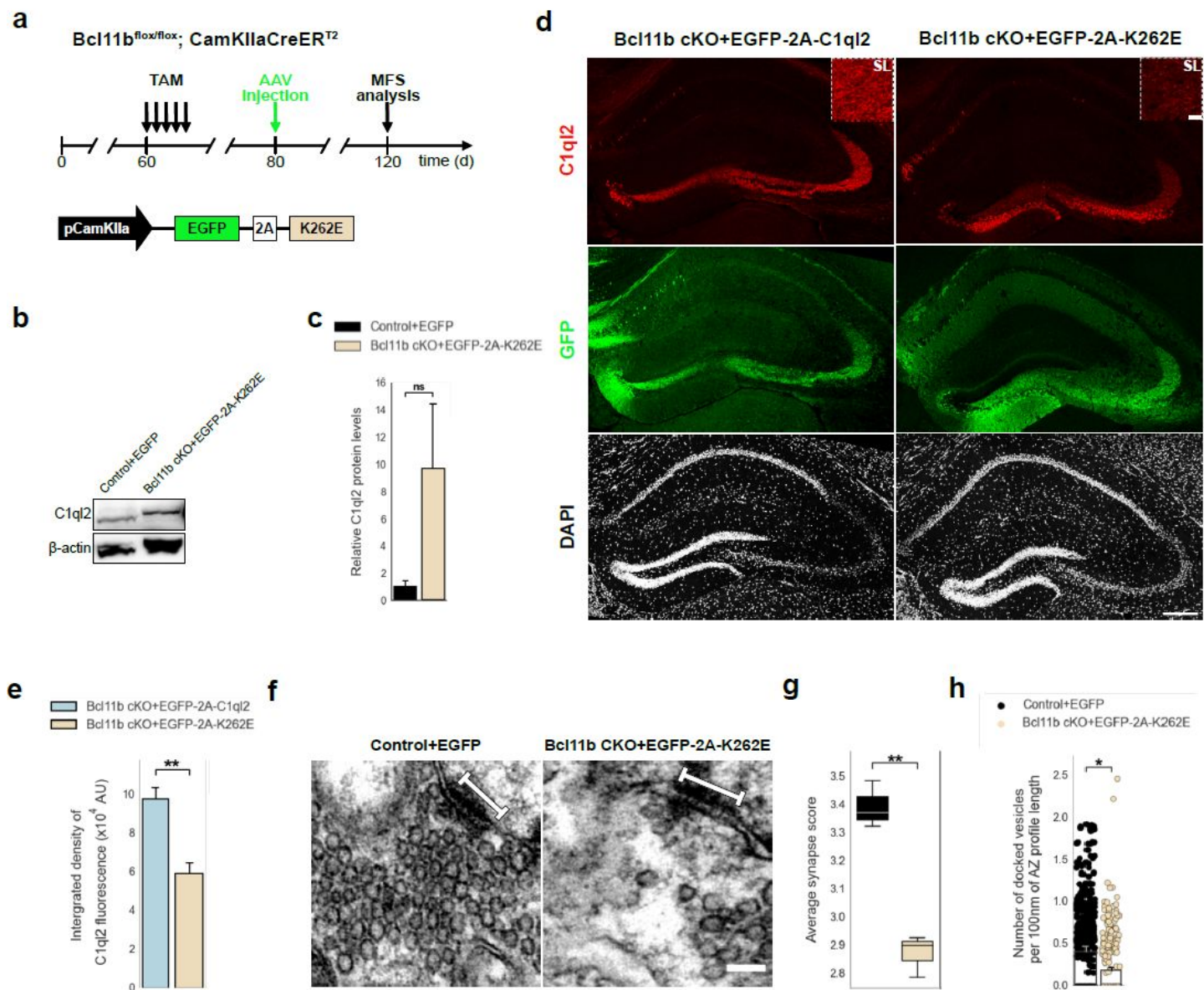


Figure 6.

C1q2-Nrxn3(25b+) interaction is important for C1q2 localization at the MFS and SV recruitment

a Experimental design to analyze the MFS after AAV-mediated expression of C1q2.K262E in *Bcl11b* cKO DGN. **b** Western blot and **c** relative C1q2.K262E protein levels in mouse hippocampal homogenates. $n=3$. All data are presented as means; error bars indicate SEM. Mann-Whitney U-test. ns: non-significant. **d** Immunohistochemistry of C1q2 (red) and GFP (green) in hippocampal sections. Scale bar: 200 μ m. Upper right corner of C1q2 panels depicts close-ups from the SL of CA3. Scale bar: 15 μ m. **e** Integrated density of C1q2 fluorescence in the SL of CA3. $n=3$. All data are presented as means; error bars indicate SEM. Unpaired t-test. $*p=0.008$. **f** Electron microscope images of MFS and proximal SV. White bars mark synapse length from postsynaptic side. Scale bar: 100 nm. **g** Average synapse score. $n=3$. Unpaired t-test. $**p=0.0012$. **h** Number of docked vesicles per 100 nm AZ profile length. $n=3$. All data are presented as means; error bars indicate SEM. Points represent the individual examined AZ. Unpaired t-test. $*p=0.018$.

To investigate how the misdistribution of C1ql2.K262E in the SL affected the structure of MFS terminals, we determined if C1ql2.K262E was able to recover the average synapse score in the *Bcl11b* cKO background. As expected, we found an ineffective activity of the mutant molecule (**Figure 6f-g**, **Supplementary figure 4b**; Control+EGFP: 3.39 ± 0.048 , *Bcl11b* cKO+EGFP-2A-K262E: 2.87 ± 0.043 , mean $\pm$ SEM), consistent with a role for C1ql2 in organizing MFS structure. Furthermore, the number of docked vesicles per 100 nm of AZ profile length in the MFB of animals receiving the C1ql2.K262E AAV was significantly lower compared to control animals (**Figure 6h**, **Supplementary figure 4c-d**; Control+EGFP: 0.41 ± 0.069 , *Bcl11b* cKO+EGFP-2A-K262E: 0.19 ± 0.035 , mean $\pm$ SEM), indicating that C1ql2-dependent signaling also affected the docking of SV within these synapses. In contrast, the diameter of docked vesicles was unchanged (**Supplementary figure 4e**; Control+EGFP: 39.06 ± 1.22 , *Bcl11b* cKO+EGFP-2A-K262E: 36.38 ± 2.19 , mean $\pm$ SEM). Unexpectedly, however, C1ql2.K262E was able to rescue the loss of MF-LTP observed in *Bcl11b* cKO (**Supplementary figure 4f-h**; 0-10 min: Control+EGFP: 90.4 ± 7.2 , *Bcl11b* cKO+EGFP-2A-K262E: 155.8 ± 30.3 , 10-20 min: Control+EGFP: 42.7 ± 3.6 , *Bcl11b* cKO+EGFP-2A-K262E: 68.7 ± 17.3 , 20-30 min: Control+EGFP: 52.5 ± 7.6 , *Bcl11b* cKO+EGFP-2A-K262E: 55.9 ± 13.8 , 30-40 min: Control+EGFP: 50.1 ± 7.3 , *Bcl11b* cKO+EGFP-2A-K262E: 47.8 ± 9.3 , mean $\pm$ SEM). Thus, our data suggest that *Bcl11b* regulates MFS function through divergent C1ql2-dependent downstream signaling pathways: while SV recruitment depends on a direct interaction of C1ql2 with *Nrxn3*(25b+), C1ql2 appears to regulate MF-LTP through *Nrxn3*(25b+)-independent mechanisms.

To explore if the binding to *Nrxn3a*(25b+) is required for correct C1ql2 targeting to the SL *in vivo*, we stereotactically injected an AAV expressing GFP-tagged Cre or inactive Cre into the DG of 2 month old *Nrxn123^{fllox/fllox}* mice (**Figure 7a**), which resulted in strong reduction of *Nrxn3* mRNA levels 2 months later, (**Figure 7b**; inactive Cre: 1 ± 0.127 , Cre: 0.38 ± 0.09 , mean $\pm$ SEM). Strikingly, endogenous C1ql2 expression along the MF axons in the SL of CA3 was significantly reduced in *Nrxn123* cKO animals compared to *Nrxn123^{fllox/fllox}* animals expressing inactive Cre (**Figure 7c-d**; inactive Cre: 11.65 ± 1.75 , Cre: 4.71 ± 0.93 , mean $\pm$ SEM). To control for the specificity of this effect, we also determined the level of C1ql3 expression and found it unchanged in *Nrxn123* cKO (**Supplementary figure 5a**). To exclude that the reduced C1ql2 expression was simply a consequence of an overall loss of MFB, we used ZnT3 as a marker of MFB and found it unchanged in *Nrxn123* cKO compared to controls (**Figure 7e**). Remarkably, disruption of the C1ql2-*Nrxn3*(25b+) binding by ablation of *Nrxn3* in *Nrxn123* cKO mutants not only led to the reduced C1ql2 expression (**Figure 7c**) but in fact recapitulated the phenotype observed upon *Bcl11b* ablation or by KD of *C1ql2* as evidenced by a large reduction of the average synapse score in *Nrxn123* cKO (**Figure 7f-g**, inactive Cre: 3.11 ± 0.06 , Cre: 2.67 ± 0.074 , mean $\pm$ SEM). Also, similarly to the *Bcl11b* and *C1ql2* mutant phenotypes, we observed the number of docked vesicles per 100 nm of AZ profile length in *Nrxn123* cKO to be diminished compared to controls (**Figure 7h**, **Supplementary figure 5c-d**; inactive Cre: 0.404 ± 0.035 , Cre: 0.197 ± 0.019 , mean $\pm$ SEM), whereas the diameter of docked vesicles remained unchanged (**Supplementary figure 5e**; inactive Cre: 38.38 ± 0.44 , Cre: 37.12 ± 0.8 , mean $\pm$ SEM). Thus, our results provide compelling evidence that *Bcl11b* controls MFS organization through C1ql2/*Nrxn3*(25b+)-dependent signaling, explicating how *Bcl11b*, a transcription factor with a broad range of functions, can regulate highly specific processes in the brain.

Discussion

There is emerging evidence that the zinc finger transcription factor *Bcl11b* is involved in the pathogenesis of neurodevelopmental as well as neuropsychiatric disorders that are frequently associated with synaptic dysfunction. Previous work from our group demonstrated *Bcl11b* to be essential for synapse function in the mossy fiber circuit of the adult murine hippocampus. The underlying molecular mechanisms downstream of *Bcl11b*, however, remained elusive. In the present study, we uncover a novel C1ql2-dependent regulatory pathway through which *Bcl11b* controls the structural as well as functional integrity of hippocampal MFS in adult mice. We show

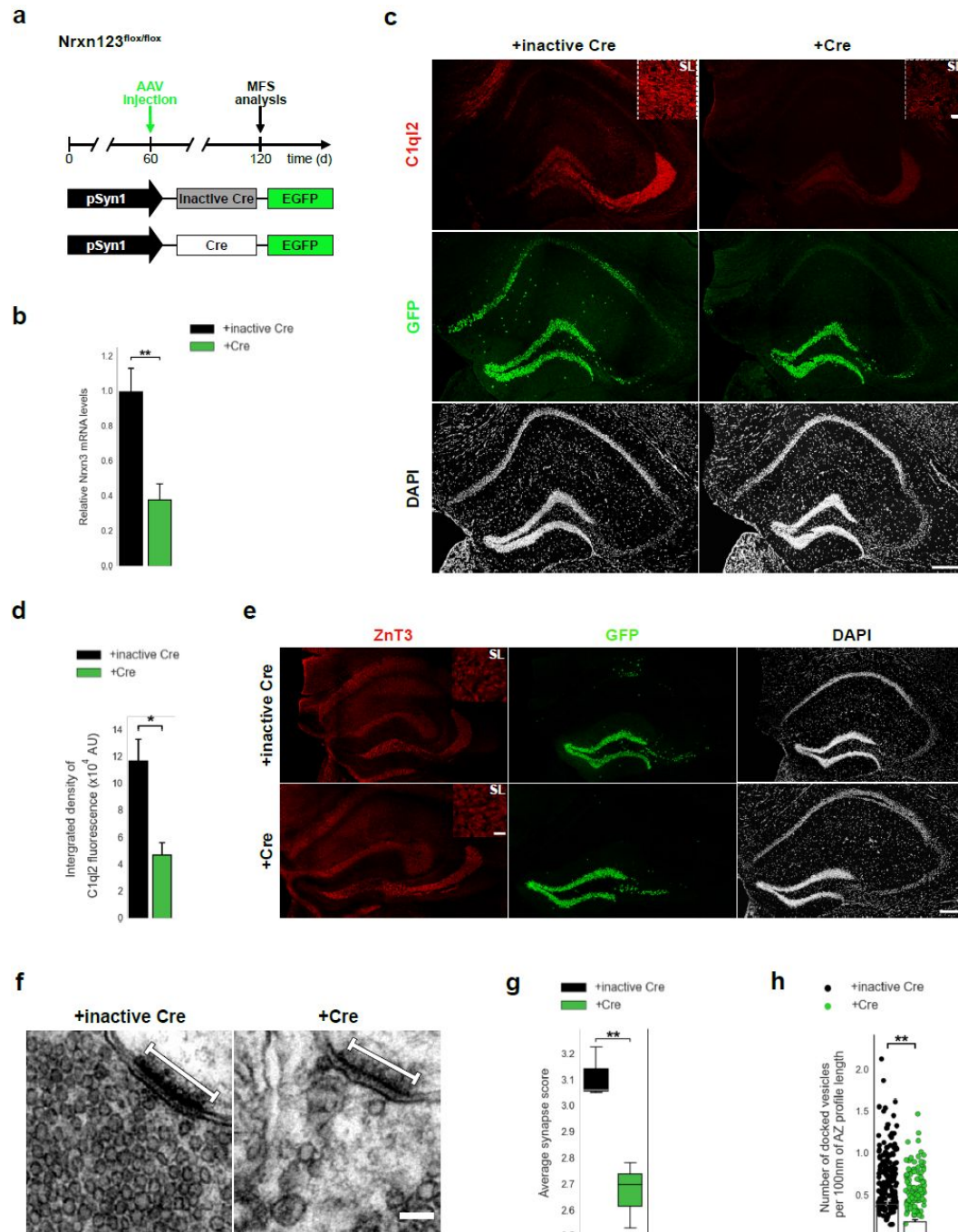


Figure 7.

Nrxn KO perturbs C1ql2 localization at the MFS and SV

a Experimental design to analyze the MFS after AAV-mediated Nrnx KO. **b** Relative Nrnx3 mRNA levels. $n=4$. All data are presented as means; error bars indicate SEM. Unpaired t-test. $^{**}p=0.007$. **c** Immunohistochemistry of C1ql2 (red) and GFP (green) in hippocampal sections. Scale bar: 200 μ m. Upper right corner of C1ql2 panels depicts close-ups from the SL of CA3. Scale bar: 15 μ m. **d** Integrated density of C1ql2 fluorescence in the SL of CA3. $n=3$. All data are presented as means; error bars indicate SEM. Unpaired t-test. $^*p=0.02$. **e** Immunohistochemistry of ZnT3 (red) and GFP (green) in hippocampal sections. Scale bar: 200 μ m. Upper right corner of ZnT3 panels depicts close-ups from the SL of CA3. Scale bar: 15 μ m. **f** Electron microscope images of MFS and proximal SVs. White bars mark synapse length from postsynaptic side. Scale bar: 100 nm. **g** Average synapse score. $n=3$. Unpaired t-test. $^{**}p=0.009$. **h** Number of docked vesicles per 100 nm AZ profile length. $n=3$. All data are presented as means; error bars indicate SEM. Points represent the individual examined AZ. Unpaired t-test. $^{**}p=0.007$.

that SV recruitment to the AZ of MFS, as well as the expression of MF-LTP, depend on C1ql2, which is a direct functional target of Bcl11b. Reintroduction of C1ql2 into *Bcl11b* mutant DGN restores defective SV recruitment and LTP expression. KD of *C1ql2* in DGN recapitulates the impaired SV recruitment and loss of LTP observed in *Bcl11b* mutants. Finally, we show that C1ql2 controls SV recruitment through a direct interaction with presynaptic Nrnx3(25b+), while LTP potentially depends on a Nrnx3-independent pathway. Recent studies suggested Nrnx3, as well as C1ql2, to be associated with neuropsychiatric disorders. Thus, our study for the first time identifies a Bcl11b/C1ql2/Nrnx3-dependent signaling pathway in the control of basic structural and functional properties of MFS. Analysis of this regulatory pathway in mice may provide important novel insights into the pathogenesis of neurodevelopmental and neuropsychiatric disorders.

We have previously shown that conditional ablation of *Bcl11b* in the adult hippocampus leads to structural and functional changes of MFS characterized by an overall reduction in synapse numbers, loss of bouton complexity, misdistribution of SV as well as loss of MF-LTP (De Bruyckere et al., 2018). Here, we found that AAV-mediated reintroduction of the synaptic organizer protein C1ql2, which is a direct transcriptional target of Bcl11b and is downregulated in *Bcl11b* mutant DGN (De Bruyckere et al., 2018), was able to rescue major part of the *Bcl11b* mutant phenotype at the MFS. Restoring C1ql2 expression in *Bcl11b* cKO DGN led to a complete rescue of the SV distribution and docking as well as LTP at the MFS, while synapse numbers and ultrastructural complexity of boutons remained unchanged. Furthermore, KD of *C1ql2* in WT DGN recapitulated the *Bcl11b* phenotype with impaired SV recruitment and loss of LTP, supporting the specificity of C1ql2 function. MF-LTP, which manifests as a long-term increase in presynaptic vesicle release probability (P_r) (Shahoha et al., 2022), directly depends on the distribution of SV at AZ. Recent studies have shown that the increase in P_r involves the recruitment of new AZ and an increase in the number of docked and tethered vesicles, corresponding to the readily releasable pool of SV (Orlando et al., 2021; Vandael et al., 2020). The perturbed SV recruitment in both *Bcl11b* cKO and *C1ql2* KD mice could thus explain the loss of LTP in both conditions. Indeed, reintroduction of C1ql2 in *Bcl11b* cKO DGN specifically rescued SV recruitment and LTP, while synapse numbers and ultrastructural complexity of MFB remained unchanged. It is important to note, however, that additional factors have been suggested to contribute to the increase in P_r , including a tighter coupling between Ca^{2+} channels and SV (Midorikawa & Sakaba, 2017) and the accumulation of Ca^{2+} channels near release sites (Fukaya et al., 2021). Therefore, it cannot be excluded that C1ql2 regulates MF-LTP through one of these alternative mechanisms. Aiming to narrow in on the nature of the mechanism through which Bcl11b regulates MF-LTP, we used forskolin to induce LTP in *Bcl11b* cKO. MF-LTP relies on presynaptic mechanisms (Castillo, 2012; Zalutsky & Nicoll, 1990), with a central role for the second messenger cAMP. This mediator is produced by AC in response to Ca^{2+} influx through voltage-gated Ca^{2+} channels (Li et al., 2007) and kainate receptors (KAR) (Lauri et al., 2001; Schmitz et al., 2003). By using forskolin to directly activate AC we bypassed these initial steps, and still found a reduction of LTP in *Bcl11b* cKO mice, similarly to HFS. These results confirm that the loss of LTP is caused by a process downstream of the initial presynaptic Ca^{2+} influx following stimulation. We note that, in the present experiments, we did not observe the decrease in input-output relation as reported in De Bruyckere et al. (2018). After excluding technical differences, e.g. different methods of data analysis, we conclude that the discrepancy is best explained by differences in the population of presynaptic fibers. In the present study, mossy fiber responses were specifically identified by testing for frequency facilitation and sensitivity to mGluR antagonists, whereas in the previous study this purification was not done (De Bruyckere et al., 2018). Together, our results suggest Bcl11b to be an important synaptic regulator that controls the structure and function of adult MFS through both C1ql2-dependent, as well as -independent transcriptional programs.

C1ql proteins are complement-related factors that are synthesized by the presynapse and secreted into the synaptic cleft. Within the hippocampus, C1ql2 and -3 protein expression overlaps and is highly restricted to DGN (Iijima et al., 2010) and the corresponding mossy fiber system, including MF-CA3 synapses. C1ql2 and -3 were previously suggested to form functional

heteromers that are able to cluster postsynaptic KAR on MFS. Selective deletion of either C1ql2 or -3 in mice was reported to have no overt mutant hippocampal phenotype, suggesting functional compensation for both proteins (Matsuda et al., 2016). Using shRNA-mediated selective KD of C1ql2 in DGN as well as rescue of the *Bcl11b* mutation by the reintroduction of C1ql2 into mutant DGN, we observed a novel presynaptic function for C1ql2 in the recruitment of SV and the expression of LTP in MFS. This function was specific to C1ql2 since overexpression of C1ql3 in *Bcl11b* mutant DGN was unable to rescue the synapse phenotype. Previous *in vitro* studies suggested that C1ql2 function at the MFS involves interaction with Nrnx3 isoforms containing the SS5^{25b} sequence (Matsuda et al., 2016). Nrnx3s are synaptic cell adhesion molecules that mediate various synaptic properties (Reissner et al., 2013; Sudhof, 2017), including the recruitment of SV and dense-core vesicles (Dean et al., 2003; Ferdos et al., 2021; Quinn et al., 2017; Rui et al., 2017). This prompted us to analyze, whether C1ql2-dependent SV recruitment in MFS requires a direct interaction with Nrnx3(25b+) *in vitro* and *in vivo*. Expression of C1ql2 in HEK293 cells co-cultured with GFP-Nrnx3α(25+)-expressing hippocampal neurons was able to recruit Nrnx3α(25b+) and vGlut1 at contact points, while C1ql2.K262E, a C1ql2 variant with an amino-acid replacement that perturbs the interaction with Nrnx3(25b+), was no longer able to recruit neuronal vGlut1. Furthermore, clustering of vGlut1 by C1ql2-expressing HEK293 cells was reduced in neurons harboring a pan-neurexin mutation, a phenotype that was rescued by the selective reintroduction of Nrnx3α(25b+). Finally, the introduction of C1ql2.K262E in *Bcl11b* cKO DGN *in vivo* was unable to rescue the SV recruitment, while the silencing of *Nrnx3* in DGN *in vivo* perturbed the SV recruitment to a similar extent as in *Bcl11b* cKO and C1ql2 KD. Based on these findings, we anticipated the overexpression of C1ql2.K262E in *Bcl11b* cKO DGN to be unable to rescue MF-LTP. Unexpectedly, the introduction of C1ql2.K262E into *Bcl11b* cKO fully rescued MF-LTP. This raises the possibility that C1ql2 can influence MF-LTP through additional, yet uncharacterized mechanisms, independent of SV recruitment or direct interaction with Nrnx3(25b+). We cannot exclude, however, that the expression of a mutant C1ql2 variant created an additional gain-of-function effect that circumvented SV recruitment and allowed the rescue of MF-LTP in our experimental system. The latter is supported by the fact that within the first 10 minutes after HFS, fEPSP slopes for C1ql2.K262E were significantly elevated compared to controls, an effect which was not seen after C1ql2 re-expression. Together, our data provide comprehensive experimental evidence that the direct interaction of C1ql2 with Nrnx3(25b+) is essential for SV recruitment at the MFS. The C1q domain is able to form stable, higher-order oligomers (Ressl et al., 2015). Neurexins, on the other hand, are highly mobile outside and inside of synaptic terminals (Klatt et al., 2021; Neupert et al., 2015). We observed that an abolished interaction between C1ql2 and Nrnx3(25b+) was accompanied by impaired targeting of the C1ql2 protein to the MFS. Thus, the interaction of C1ql2 with Nrnx3(25b+) may reciprocally augment the accumulation of both proteins at synaptic sites.

Perturbations in synaptic structure and function are major determinants of various neuropsychiatric and neurodevelopmental disorders (Hayashi-Takagi, 2017; Lepeta et al., 2016; Zoghbi & Bear, 2012). Emerging evidence from recent genetic studies suggests such disorders to be linked to various genes encoding for synaptic proteins (Sudhof, 2021; Torres et al., 2017; Wang et al., 2018). Decoding the molecular mechanisms of synaptic organization and stability and their transcriptional regulation would therefore be expected to contribute to the mechanistic understanding of neuropsychiatric and neurodevelopmental disorders. The transcription factor *Bcl11b* has been linked to neurodevelopmental (Lessel et al., 2018), neurodegenerative (Kunkle et al., 2016; Song et al., 2022) and neuropsychiatric disorders (Whitton et al., 2018; Whitton et al., 2016). *BCL11B* mutations in humans are associated with neurodevelopmental delay, overall learning deficits as well as impaired speech acquisition and autistic features (Eto et al., 2022; Lessel et al., 2018; Punwani et al., 2016; Yang et al., 2020). Moreover, conditional ablation of *Bcl11b* selectively in the adult murine hippocampus results in impaired learning and memory behavior (Simon et al., 2016). *NRXN3* single-nucleotide polymorphisms (SNP) have been implicated in schizophrenia (Hu et al., 2013) and addiction (Hishimoto et al., 2007), with one recorded SNP located close to SS5 altering the

expression of *Nrxn3*(25b+). Interestingly, recent studies have also associated *C1QL2* with schizophrenia (Marballi et al., 2022 [DOI](#)) as well as cocaine addiction (Huggett & Stallings, 2020a [DOI](#), 2020b [DOI](#)). In this study we demonstrate that *Bcl11b*, through its transcriptional target *C1ql2*, modulates the synaptic organization of MFS by controlling the recruitment of SV at AZ. This regulatory mechanism depends on a direct interaction of *C1ql2* with *Nrxn3*(25b+). Importantly, SV trafficking and altered release probability have been implicated in neurological and neuropsychiatric disorders (Egbujo et al., 2016 [DOI](#); Lepeta et al., 2016 [DOI](#); Zhu et al., 2021 [DOI](#)). Thus, the identification of the *Bcl11b*/*C1ql2*/*Nrxn3*(25b+)-dependent signaling module in this study provides a new entry point for future mechanistic analyses of synaptopathies. Moreover, the existence of such cell type-specific signaling modules reveals how a fundamental transcription factor with diverse functions such as *Bcl11b* can be implicated in the pathomegenesis of brain disorders characterized by synaptic dysfunction.

Materials and Methods

Animals

Bcl11b inducible mutants were generated as previously described (De Bruyckere et al., 2018 [DOI](#)). *Bcl11b*^{flox/flox}; CaMKIIa-CreER^{T2} (*Bcl11b* cKO) and *Bcl11b*^{+/+}; CaMKIIa-CreER^{T2} (control) littermates were used. The *Bcl11b* mutation was induced by intraperitoneal injection of 2 mg tamoxifen for five consecutive days. C57BL/6J mice were obtained from Janvier-Labs. For the pan-*neurexin* KO, *Nrxn123*^{flox} mice (Chen et al., 2017 [DOI](#)) were used. Animals were kept in a 12:12-h light-dark cycle at a constant temperature (22 ± 1 °C) in IVC cages. All mouse experiments were carried out in compliance with the German law and approved by the respective government offices in Tübingen and Karlsruhe, Germany.

Stereotaxic Injections

For the expression of *C1ql2* and *C1ql3*, the DG of 80 days old *Bcl11b* cKO mice were injected with AAV vectors expressing EGFP-2A-*C1ql2* and EGFP-2A-*C1ql3*, respectively. As control, *Bcl11b* cKO and control mice were injected with an AAV expressing EGFP. For the KD of *C1ql2*, the DG of 60 days old C57BL/6J mice were injected with AAV 4xsh*C1ql2*-EGFP, expressing 4 shRNAs against *C1ql2* or control AAV 4xshNS-EGFP, expressing 4 non-sense shRNAs. For pan-*neurexin* KO, the DG of 60 days old *Nrxn123*^{flox/flox} mice were injected with an AAV expressing EGFP-Cre or a control AAV expressing EGFP-Cre.Y324F, an inactive Cre. All AAV were produced by the Viral Vector Facility of the Neuroscience Center Zurich on request. The mice were anesthetized with 5% isoflurane and placed in a mouse stereotaxic apparatus. During the entire procedure, anesthesia was maintained by constant administration of 2.2% isoflurane. Eye ointment was applied to prevent eyes from drying. For electrophysiological experiments, mice were subcutaneously injected with buprenorphine hydrochloride (0.1 mg/kg, Temgesic, Indivior) 30 minutes before and 3 hours after each surgery. For all other experiments, Butorphanol (Livisto) and Meloxicam (Boehringer-Ingelheim; 5 µg/g) were injected subcutaneously and the local anesthetic Bupivacaine (Puren; 5 µg/g) was injected subcutaneously at the incision site. After 10 min the head of the mouse was shaved and disinfected and an incision was made in the skin. Targeted injection sites were identified and a small craniotomy was performed for each site. The injector was placed at the individual sites and the viral solution was injected at 100 nL/min, with a 5-10 min recovery before removing the injector. After injections at all sites the incision was sutured and the animal was monitored for recovery from anesthesia, after which it was returned to its home cage. For histological and EM analyses of MFS, AAV were injected at three sites per hemisphere with the following coordinates (Bregma: AP 0; ML: 0; DV:0): AP -2 mm; ML ±1 mm; DV -2 mm. AP -2.5 mm; ML ±1.5 mm; DV -1.8 mm. AP -3.1 mm; ML ±2, DV -2.2 mm. For electrophysiological analyses, AAV were injected at two dorsoventral coordinates per hemisphere: AP -3.0 mm; ML ±3.25 mm; DV -2.4 and -2.8 mm. 200-300 nL of AAV (1e12 vg/mL) were injected in each location.

RNA isolation and quantitative real-time PCR

All procedures were performed in an RNase-free environment. Animals were sacrificed under deep CO₂-induced anesthesia, brains were quickly dissected in ice-cold PBS, cryopreserved in 20% sucrose overnight, frozen in OCT compound (Polysciences), and stored at -80 °C. 20 µm thick coronal sections were collected on UV-treated and 0.05% poly-L-lysine coated membrane-covered PEN slides (Zeiss), fixed for 1 min in ice-cold 70% EtOH, incubated for 45 sec in 1% cresyl violet acetate solution (Waldeck) and washed for 1 min each in 70% EtOH and 100% EtOH. Sections were briefly dried on a 37 °C warming plate and immediately processed. The granule cell layer of the DG was isolated by laser capture microdissection using a PALM MicroBeam Rel.4.2 (Zeiss). RNA was isolated from the collected tissue using RNeasy Micro Kit (Qiagen) and reverse transcribed using the SensiFast cDNA Synthesis Kit (Bioline). Quantitative real-time PCR was performed in triplets for each sample using the LightCycler DNA Master SYBR Green I Kit in a LightCycler 480 System (Roche). The relative copy number of Gapdh RNA was used for normalization. Data were analyzed using the comparative CT method (Schmittgen & Livak, 2008 [\[3\]](#)).

Western blots

Briefly, hippocampi from freshly removed brains were dissected in ice-cold PBS, collected in Lysis Buffer (50mM Tris pH 7.5, 150mM NaCl, 0.5% sodium deoxycholate, 1% triton-X100, 0.1% SDS) and manually homogenated. Samples were centrifuged for 25 min at 13200 rpm at 4 °C and the supernatant was collected. Protein concentration was calculated with Bradford assay. Protein suspension containing 40 µg of protein was mixed 1:1 with 2x SDS loading dye (62.5mM Tris, 10% Glycerol, 5% β-mercaptoethanol, 80mM SDS, 1.5mM bromophenol blue), boiled at 95 °C for 5 min, separated by SDS-PAGE and electrophoretically transferred onto PVDF membranes (Merck). Membranes were blocked with 5% non-fat milk (Sigma-Aldrich), incubated with mouse anti-β-actin (1:5000; Sigma-Aldrich) and rabbit anti-C1ql2 (1:500; Sigma-Aldrich), followed by Peroxidase-conjugated secondary antibodies (Jackson ImmunoResearch) and developed with Pierce ECL Western Blotting Substrate (ThermoScientific). Signal was detected with ChemiDoc Imaging System (BioRad) and analyzed with Image Lab Software (BioRad). Protein signal was normalized with the signal of β-actin.

Immunohistochemistry and RNA in situ Hybridization

Animals were sacrificed under deep CO₂-induced anesthesia, brains were dissected in ice-cold 1x PBS, and either fixed for 4 h in 4% PFA in PBS at 4 °C and cryopreserved in 20% sucrose in PBS overnight at 4 °C or directly cryopreserved and then frozen in OCT compound (Polysciences). Sections were prepared at 14 µm. The unfixed sections were postfixated with 4% PFA in 1x PBS for 20 min. Heat-induced antigen retrieval in 10mM citrate buffer (pH 6.0) was performed for fixed sections. Sections were blocked at RT for 1 h in 1x PBS containing 0.1% TritonX-100 and 10% horse serum, and incubated overnight at 4 °C with primary antibodies, followed by a 90 min incubation with secondary antibodies. Sections were counterstained with DAPI (Molecular Probes). The following primary antibodies were used on fixed sections: guinea pig anti-Bcl11b (1:1000; [Simon et al., 2012 \[3\]](#)), rabbit anti-C1ql2 (1:1000; Invitrogen), rabbit anti-C1ql2 (1:500; Sigma-Aldrich), chicken anti-GFP (1:2000; Abcam) and rabbit anti-C1ql3 (1:500; Biozol). Primary antibodies used on unfixed sections: mouse anti-vGlut1 (1:100; Synaptic Systems), guinea pig anti-Homer1 (1:250; Synaptic Systems), rabbit anti-C1ql2 (1:1000; Invitrogen) and rabbit anti-ZnT3 (1:200; Synaptic Systems). All fluorescent secondary antibodies were purchased from Jackson ImmunoResearch and used at 1:500 dilution. Hybridizations were performed with DIG-labelled riboprobes on 14 µm thick sections.

Transmission electron microscopy

Animals were sacrificed through CO₂-inhalation and immediately perfused transcardially with 0.9% NaCl for 1 min, followed by a fixative solution of 1.5% glutaraldehyde (Carl Roth) and 4% PFA in 0.1M PB pH 7.2 for 13 min. Brains were dissected and postfixated in the fixative solution for 4 h at

4 °C. Ultrathin sections (60 nm) were prepared and stained with lead citrate. Images were acquired using a transmission electron microscope LEO 906 (Zeiss) with a sharp-eye 2k CCD camera and processed with ImageSP (Tröndle). Synapse score (De Bruyckere et al., 2018 [DOI](#)) was calculated according to the following criteria: 0-5 vesicles above the active zone = 0; 5-20 vesicles = 1; small group of vesicles ($\leq 200000 \text{ nm}^2$) with distance between density and closest vesicle $> 100 \text{ nm}$ = 2; small group of vesicles ($\leq 200000 \text{ nm}^2$) with distance between density and closest vesicle $\leq 100 \text{ nm}$ = 3; big group of vesicles ($> 200000 \text{ nm}^2$) with distance between density and closest vesicle ($> 100 \text{ nm}$) = 4; big group of vesicles ($> 200000 \text{ nm}^2$) with distance between density and closest vesicle $\leq 100 \text{ nm}$ = 5. Synapses from approximately 30 MFB per animal were analyzed. Vesicles with a distance $\leq 5 \text{ nm}$ from the plasma membrane were considered docked (Kusick et al., 2022 [DOI](#); Vandael et al., 2020 [DOI](#)). Approximately 100 AZ per animal were analyzed.

Electrophysiological recordings and data analysis

Animals were sacrificed under deep CO_2 -induced anesthesia at 4 months. Brains were quickly removed and placed in ice-cold modified ACSF containing (in mM) 92 N-methyl-D-glucamine (NMDG), 2.5 KCl, 1.2 NaH_2PO_4 , 30 NaHCO_3 , 20 HEPES, 25 glucose, 5 Na-ascorbate, 2 thiourea, 3 Na-pyruvate, 10 MgSO_4 , 0.5 CaCl_2 , 6 N-acetyl-L-cysteine (NAC), saturated with carbogen gas (95% O_2 and 5% CO_2 , pH 7.4) (Ting et al., 2014 [DOI](#)). 450 μm thick horizontal slices were cut using a vibratome slicer (Leica) at a defined angle to improve the preservation of mossy fibers (Bischofberger et al., 2006 [DOI](#)). After cutting, slices were transferred to a “Haas”-type interface chamber (Haas et al., 1979 [DOI](#)), where they were perfused with carbogen-saturated ACSF containing (in mM) 124 NaCl, 3 KCl, 2.3 CaCl_2 , 1.8 MgSO_4 , 10 glucose, 1.25 NaH_2PO_4 , 26 NaHCO_3 (pH 7.4 at 34°C) at a rate of 1.5 mL/min at $34 \pm 1^\circ \text{C}$. Slices were allowed to recover for a minimum of 1 h before the start of recordings.

Recordings were carried out by placing a glass micropipette (tip diameter 3–5 μm) filled with ACSF in the SL of the CA3b area. To induce MF field excitatory post-synaptic potentials (fEPSP), a bipolar electrode (Science Products) was placed within the hilus region of the DG. 0.1 ms pulses were delivered with an Iso-Flex stimulus isolator (AMPI) at 20 sec intervals. Putative mossy fiber signals were preliminarily identified using a 25 Hz train of five pulses. Input-output relationships were obtained by measuring the fiber volley amplitude and fEPSP slope in response to stimulations with intensities ranging from 3 to 40 V. For LTP recordings, stimulation intensity for each slice was adjusted to obtain a slope value of 20% (30% in the case of forskolin (Biomol) experiments) of the maximum fEPSP slope. LTP was induced by three trains of 100 stimulation pulses at 100 Hz (high frequency stimulation, HFS), repeated every 8 sec. 3 μM DCG-IV (Tocris Bioscience) was applied after each experiment, and only recordings displaying $>70\%$ reduction in putative MF-fEPSP slopes were used for analysis. fEPSPs were amplified 100x with an EXT 10-2F amplifier (npi electronics). Signals were low-pass filtered at 2 kHz and high-pass filtered at 0.3 Hz, digitized at 20 kHz with an analog-to-digital converter (Cambridge Electronic Design (CED)) and stored for offline analysis using Spike2 (v7) software (CED). Offline data analysis was performed on raw traces using Spike2. Slope values were measured from the linear part of the fEPSP rising phase by manually placing vertical cursors. Changes in fEPSP slopes were calculated as a percentage of the average baseline fEPSP ((average fEPSP slope in a given time interval after HFS - average fEPSP slope before HFS)/ (average fEPSP slope before HFS)).

DNA constructs

For expression, *C1ql2* was cloned from mouse cDNA. A 6xHis-myc tag was attached to the N-terminus and the construct was cloned into the pSecTag2A vector (Invitrogen) in frame with the N-terminal IgK signal sequence. A stop codon was introduced directly after *C1ql2*. The K262E point mutation was introduced with the Q5 Site-Directed Mutagenesis Kit (New England Biolabs). pSecTag2A was used for control experiments. Rat *Nrxn3a(25b+)* cDNA (Ushkaryov & Sudhof, 1993 [DOI](#)) was inserted into an pSyn5 vector with human Synapsin promoter (Neupert et al.,

2015 [\[link\]](#)) using BamHI and BglII. For the pan-*neurexin* KO and the control experiments, vectors with NLS-GFP-Cre or NLS-GFP-Cre.Y324F were used (Klatt et al., 2021 [\[link\]](#); Wang et al., 2016 [\[link\]](#)). All vectors were validated by sequencing (Eurofins Genomics).

Primary Hippocampal Cultures

Hippocampi were dissected from P0 mice in HBSS media, digested for 15 min with HBSS containing 0.1% Trypsin (Gibco) at 37 °C, dissociated in plating media (MEM supplemented with 0.6% glucose, 10% FBS, 1% penicillin/streptomycin, DNaseI 4 U/mL) and seeded on poly-L-Lysin precoated coverslips placed inside 12-well plates at 1.5×10^5 cells/mL. After 3 h the plating media was replaced with neuronal growth media (Neurobasal A supplemented with 2% B27, 2mM L-Glutamine, 1% penicillin/streptomycin, 1% N2 and 0.005% NGF). Cultures were kept at 37 °C under 5% CO₂ atmosphere. The day of plating was considered as 0 days in vitro (DIV). At DIV3 and DIV7 80% of the medium was exchanged with fresh growth medium. At DIV9 the medium was exchanged with penicillin/streptomycin-free growth medium and at DIV10 neurons were transfected using Lipofectamine 2000 (Invitrogen). Briefly, a total of 200 µL transfection mix per well was prepared by first mixing 100 µL Opti-MEM with 4 µL Lipofectamine 2000 in one tube and 100 µL Opti-MEM with 3 µg DNA in a different tube. After 5 min both volumes were combined and the mixture was incubated for 20 min at RT. The transfection mix was then added dropwise to the neurons. After 3 h of incubation the medium was exchanged with fresh growth medium.

HEK293 Cell Culture

Human embryonic Kidney (HEK) 293 cells were obtained from ATCC and were maintained in DMEM supplemented with 10% fetal calf serum and 1% penicillin/streptomycin at 37 °C under 5% CO₂ atmosphere. Cells were transfected using Lipofectamine 2000 according to the manufacturer's instructions on the same day the neurons were transfected. Cells were incubated for at least 24 h before being used in co-culture experiments.

Neuronal and HEK293 co-culture and immunostaining

Transfected HEK293 cells were washed, dissociated and resuspended in neuronal growth medium. 15×10^3 cells were added in each well containing DIV11 transfected neurons. HEK293 cells were co-cultured with the hippocampal primary neurons 2 days (DIV13 for neurons) before proceeding with immunostaining. Coverslips with cultured neurons and HEK293 cells were first fixed with 4% PFA in 1x PBS for 10 min at 4 °C, then washed 3x with 1 mL 1x PBS and blocked with 1x PBS containing 0.1% Triton X-100 and 10% horse serum for 1 h at RT. Primary antibodies were incubated overnight at 4 °C, followed by a 90 min incubation with secondary antibodies. Cells were counterstained with DAPI (Molecular Probes). The following primary antibodies were used: rabbit anti-myc-tag (1:2000; Abcam), guinea pig anti-vGlut1 (1:250; Synaptic Systems), chicken anti-GFP (1:2000; Abcam). All fluorescent secondary antibodies were purchased from Jackson ImmunoResearch and used at 1:500 dilution. For each condition, 25 cells per experiment were analyzed.

Structural protein modelling

The crystal structure of trimeric C1q-domains of mouse C1ql2 (Ressl et al., 2015 [\[link\]](#)) was used to predict a potential electrostatic binding site to splice insert 25 of Nr3n3a. An electrostatic surface map of the trimer was calculated using APBS (Jurrus et al., 2018 [\[link\]](#)). The K262E mutation was introduced using FoldX (foldxsuite.crg.eu) and was chosen in order to generate a negatively charged surface that would potentially be repulsive to Nr3n3a binding. Final models were visualized with PyMOL (*pymol.org* (<http://pymol.org/>) [\[link\]](#)).

Image acquisition and analysis

All fluorescent images of sectioned hippocampal tissue were examined on a TCS SP5II confocal microscope (Leica) using LAS-X software and processed with Fiji (Schindelin et al., 2012 [DOI](#)). Overview images were acquired with a 20x objective. Synapse numbers were quantified in the SL imaged with a 40x objective at x2 zoom. C1ql2 fluorescence intensity was quantified in the SL imaged with a 40x objective. Acquisition settings were kept constant for every sample and condition. All fluorescent images of co-cultured HEK293 cells were examined on a TCS SP8 confocal microscope (Leica) using LAS-X software and processed with Fiji. Images were acquired with a 40x objective at 4x zoom. As before, acquisition settings were kept constant for every sample and condition. Images were analyzed by masking transfected HEK293 cells and measuring the area of each mask covered by the chosen stain.

Quantification and statistical analysis

Data analysis was conducted blind to the experimental treatment. Statistical analysis and graph generation was done using Python 3. If samples met the criteria for normality, we used two-tailed unpaired t-test to compare two groups and one-way ANOVA for more than two groups. For non-normally distributed data Mann-Whitney u-test was used. Two-way ANOVA was used for examining the influence of two different categorical independent variables. If ANOVAs were significant, we used a post hoc Tukey's multiple-comparisons test to compare groups (structural and expression data) or a post hoc Bonferonni's comparison test (electrophysiological data). Data are presented as mean±SEM. Significance levels were set as indicated in figures: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Data availability

All relevant data are available from the authors.

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Author Contributions

AK, SB and AD designed the study and wrote the manuscript with input from MR, BH, PL and MM. AK, SN and BH carried out electron microscopic analyses. MR and AE carried out electrophysiological experiments. CR and MM provided neurexin cKO mice and helped with in silico modeling of C1ql2 protein interaction sites. AK generated all AAV-dependent expression constructs and carried out structural and molecular phenotype analyses of mouse mutants and cell culture experiments. EDB quantified synapse numbers. SB and AD supervised the project.

Competing Interest Statement

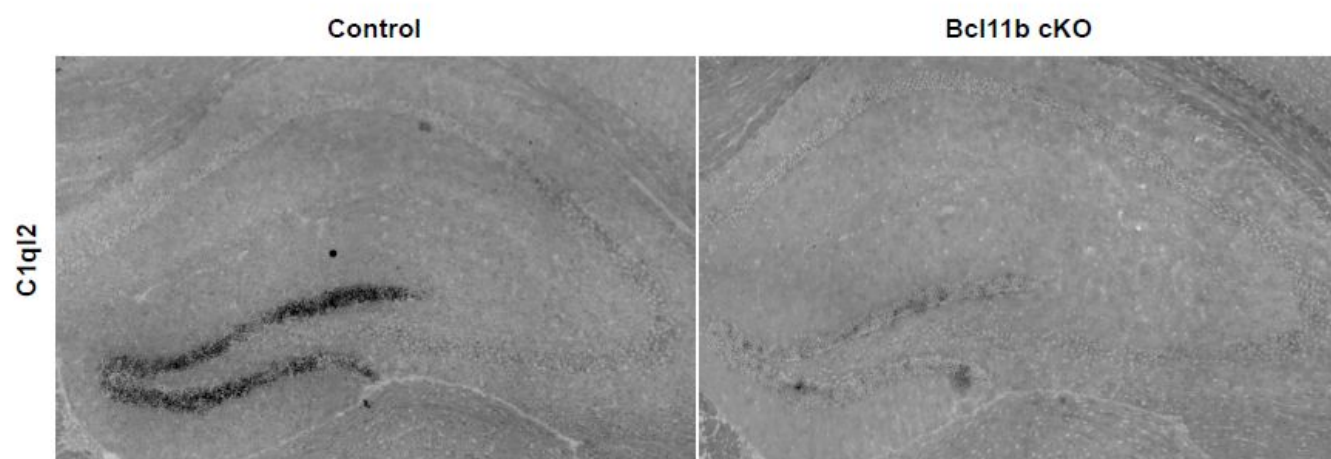
The authors declare no competing interests.

Materials & Correspondence

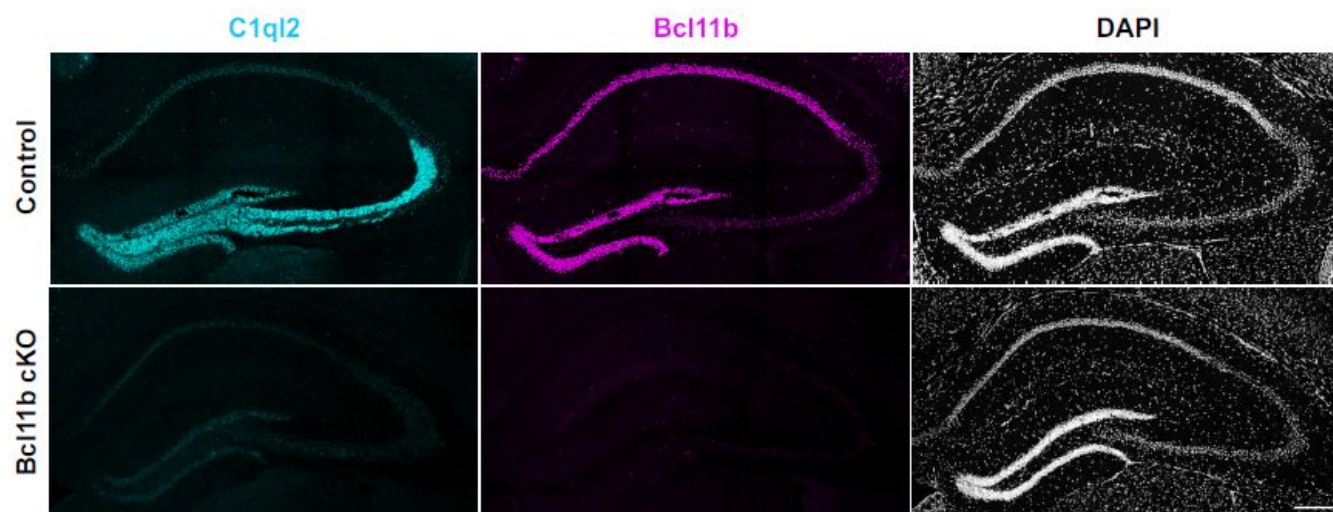
Correspondence and requests for materials should be addressed to SB.

Supplementary Data to

a



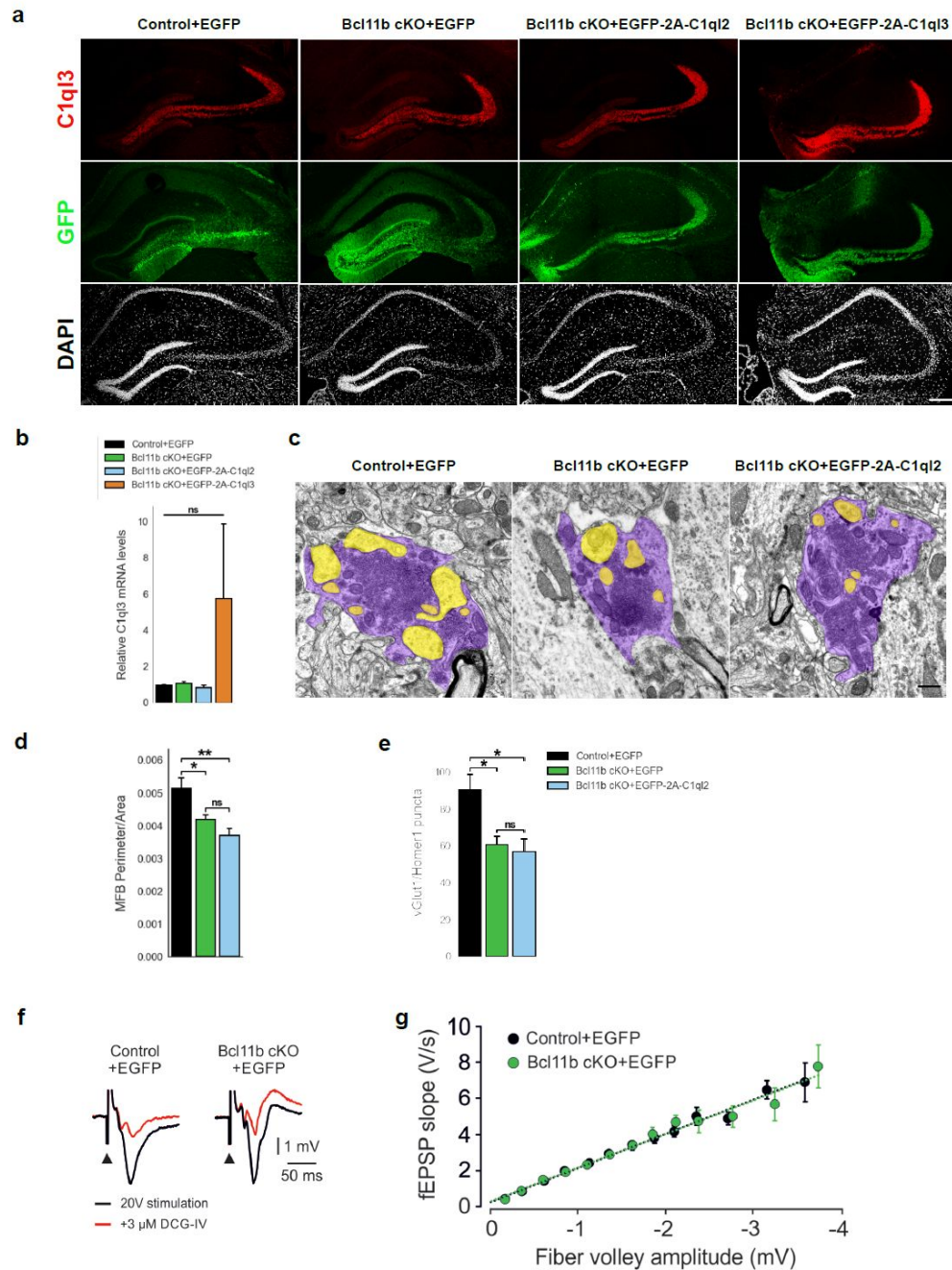
b



Supplementary Figure 1.

C1ql2 mRNA and protein are lost upon *Bcl11b* cKO in DGN

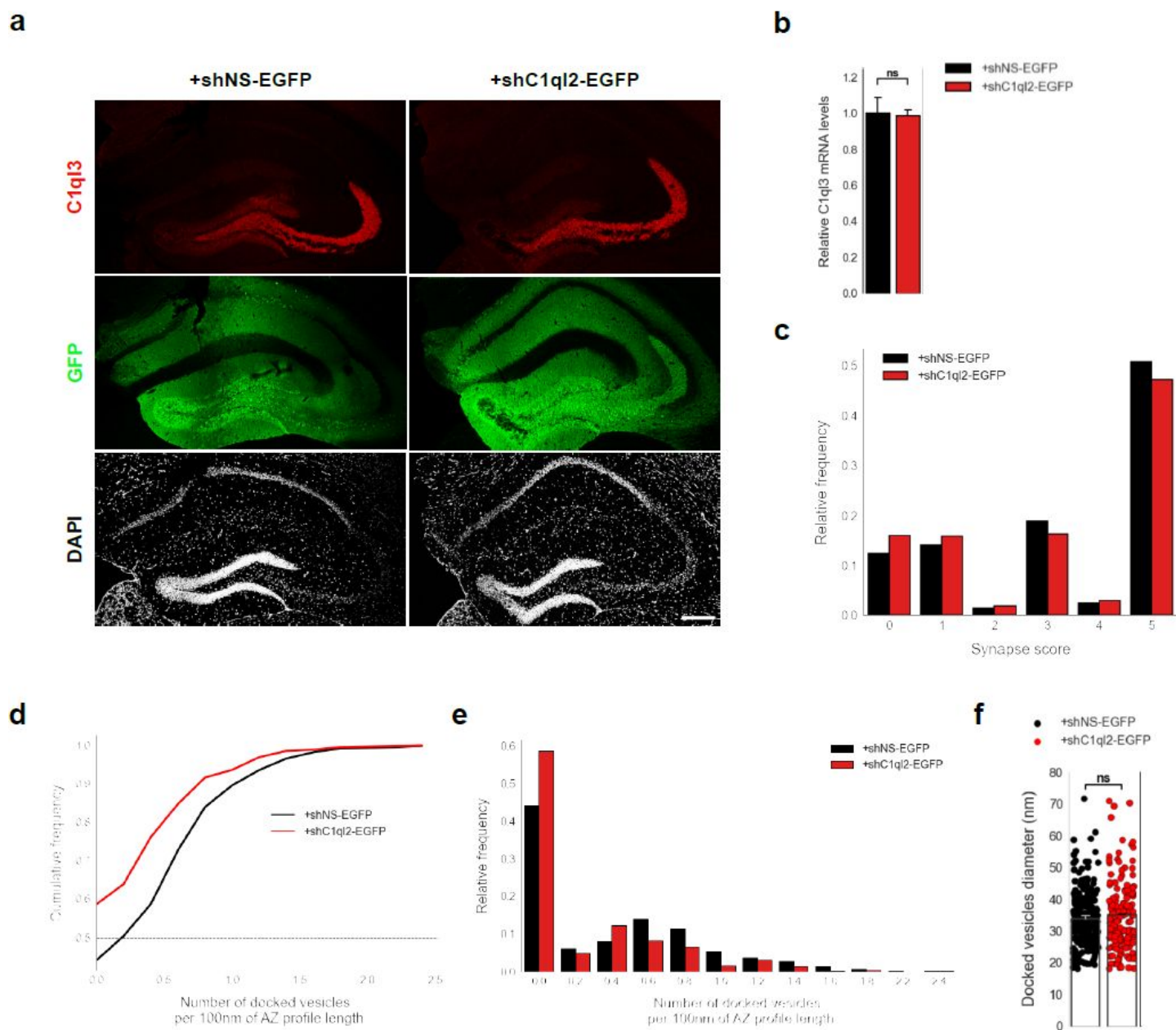
a mRNA in situ hybridization of C1ql2 on hippocampal sections. **b** Immunohistochemistry of C1ql2 (cyan) and Bcl11b (magenta) on hippocampal sections.



Supplementary Figure 2.

C1ql2 reintroduction in *Bcl11b* cKO DGN does not rescue MFS number and MFB complexity

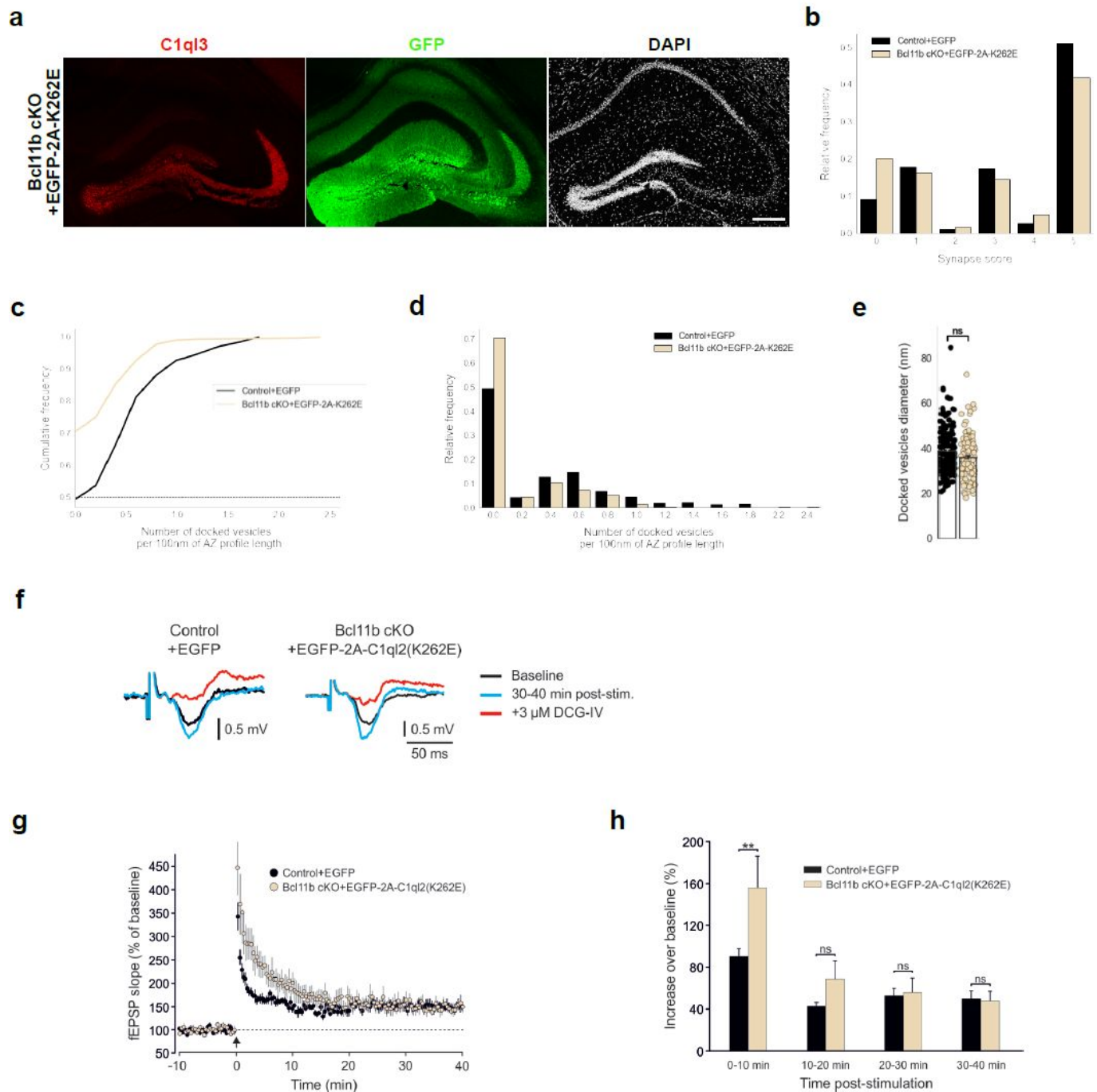
a Immunohistochemistry of C1ql3 (red) and GFP (green) on hippocampal sections. Scale bar: 200 μ m. **b** Relative *C1ql3* mRNA levels in DGN. Control+EGFP, Bcl11b cKO+EGFP, Bcl11b cKO+EGFP-2A-C1ql2, n=4; Bcl11b cKO+EGFP-2A-C1ql3, n=3. All data are presented as means; error bars indicate SEM. Two-way ANOVA. ns, not significant. **c** Electron microscopy images of MFJs (purple) and contacting postsynaptic spines (yellow). Scale bar: 500 nm. **d** MFB perimeter-to-area ratio. n=5. All data are presented as means; error bars indicate SEM. Two-way ANOVA and Tukey's PHC. Control+EGFP vs. Bcl11b cKO+EGFP: *p=0.035, and vs. Bcl11b cKO+EGFP-2A-C1ql2: **p=0.0014; ns, not significant. **e** vGlut1 and Homer1 double positive puncta in selected CA3 SL ROIs. n=3. All data are presented as means; error bars indicate SEM. Two-way ANOVA and Tukey's PHC. Control+EGFP vs. Bcl11b cKO+EGFP: *p=0.047, and vs. Bcl11b cKO+EGFP-2A-C1ql2: *p=0.029; ns, not significant. **f** Representative fEPSP traces showing responses to a 20 V electrical stimulation (black, black arrowheads indicate stimulation). The signal is almost entirely blocked by the application of 3 μ M DCG-IV (red). **g** Input-output curves generated by plotting fEPSP slope against fiber volley amplitude at increasing stimulation intensities. Control+EGFP, n = 35; Bcl11b cKO+EGFP, n = 32. The data are presented as means, error bars represent SEM.



Supplementary Figure 3.

KD of *C1ql2* in DGN of WT mice impairs SV recruitment

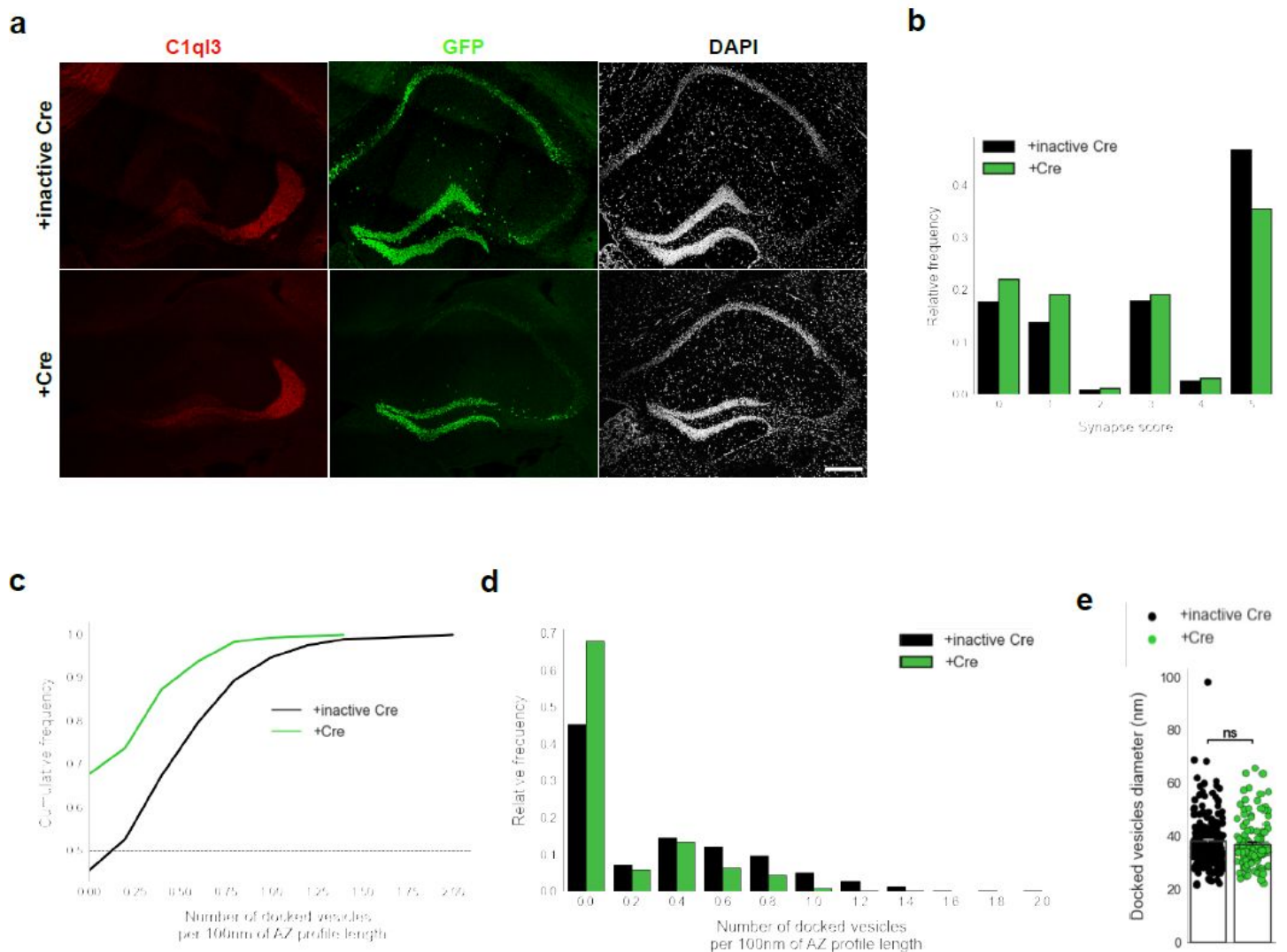
a Immunohistochemistry of *C1ql3* (red) and GFP (green) on hippocampal sections. Scale bar: 200µm. **b** Relative *C1ql3* mRNA levels in DGN. $n=4$. All data are presented as means; error bars indicate SEM. Unpaired t-test. ns, not significant. **c** Relative frequency of synapse scores (refers to [Figure 4e](#)). **d** Cumulative and **e** relative frequency of the number of docked vesicles per 100 nm AZ profile length (refer to [Figure 4f](#)). **f** Diameter of the docked vesicles. $n=3$. All data are presented as means; error bars indicate SEM. Points represent the individual examined SV. Unpaired t-test. ns, not significant.



Supplementary Figure 4.

C1q12-Nrxn3(25b+) interaction is important for SV recruitment at the MFS

a Immunohistochemistry for C1q13 (red) and GFP (green) on hippocampal sections. Scale bar: 200 μ m. **b** Relative frequency of synapse scores (refers to **Figure 6g**). **c** Cumulative and **d** relative frequency of the number of docked vesicles per 100 nm AZ profile length (refer to **Figure 6h**). **e** Diameter of the docked vesicles. n=3. All data are presented as means; error bars indicate SEM. Points represent the individual examined SV. Unpaired t-test. ns, not significant. **f** Representative fEPSP traces showing baselines before HFS (black), fEPSP changes 30-40 min after HFS (cyan) and following the application of 3 μ M DCG-IV (red). **g** Time course of fEPSP slopes. The black arrow indicates HFS and the dashed line the baseline level. **h** Quantification of fEPSP facilitation at four different time intervals after HFS. Changes in the fEPSP slope are shown as percentage of the mean baseline fEPSP. Data for C1q12.K262E from f-h in this figure are compared with control data from **Figure 3a-c**. Control+EGFP, n=7; Bcl11b cKO+EGFP-2A-K262E, n=5. All data are presented as means; error bars indicate SEM. Mann-Whitney U-test for each time interval. 0-10 min: **p=0.005; ns, not significant.



Supplementary Figure 5.

Nrxn KO perturbs C1ql2 localization and SV recruitment at the MFS

a Immunohistochemistry for C1ql3 (red) and GFP (green) on hippocampal sections. Scale bar: 200 μ m. **b** Relative frequency of synapse scores (refers to **Figure 7g**). **c** Cumulative and **d** relative frequency of the number of docked vesicles per 100 nm AZ profile length (refer to **Figure 7h**). **e** Diameter of the docked vesicles. $n=3$. All data are presented as means; error bars indicate SEM. Points represent the individual examined SV. Unpaired t-test. ns, not significant.

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Author information

Artemis Koumoundourou

Institute of Molecular and Cellular Anatomy, Ulm University, Ulm, Germany
ORCID iD: [0000-0002-8917-5717](https://orcid.org/0000-0002-8917-5717)

Märt Rannap

Institute for Physiology and Pathophysiology, Heidelberg University, Heidelberg, Germany

Elodie De Bruyckere

Institute of Molecular and Cellular Anatomy, Ulm University, Ulm, Germany

Sigrun Nestel

Department of Neuroanatomy, Institute of Anatomy and Cell Biology, Faculty of Medicine,
University of Freiburg, Freiburg, Germany

Carsten Reißner

Institute of Anatomy and Molecular Neurobiology, University of Münster, Münster, Germany

Alexei V. Egorov

Institute for Physiology and Pathophysiology, Heidelberg University, Heidelberg, Germany

Pentao Liu

School of Biomedical Sciences, Li Ka Shing Faculty of Medicine, The University of Hong Kong,
Hong Kong. Centre for Translational Stem Cell Biology, 17W, Science Park, Hong Kong

Markus Missler

Institute of Anatomy and Molecular Neurobiology, University of Münster, Münster, Germany
ORCID iD: [0000-0001-8008-984X](https://orcid.org/0000-0001-8008-984X)

Bernd Heimrich

Department of Neuroanatomy, Institute of Anatomy and Cell Biology, Faculty of Medicine,
University of Freiburg, Freiburg, Germany

Andreas Draguhn

Institute for Physiology and Pathophysiology, Heidelberg University, Heidelberg, Germany
ORCID iD: [0000-0002-6243-5582](https://orcid.org/0000-0002-6243-5582)

Stefan Britsch

Institute of Molecular and Cellular Anatomy, Ulm University, Ulm, Germany

For correspondence: stefan.britsch@uni-ulm.de

ORCID iD: [0000-0003-4379-3322](https://orcid.org/0000-0003-4379-3322)

Editors

Reviewing Editor

Jun Ding

Stanford University, United States of America

Senior Editor

Lu Chen

Stanford University, United States of America

Reviewer #1 (Public Review):

Koumoundourou et al., identify a pathway downstream of Bcl11b that controls synapse morphology and plasticity of hippocampal mossy fiber synapses. Using an elegant combination of in vivo, ex vivo, and in vitro approaches, the authors build on their previous work that indicated C1ql2 as a functional target of Bcl11b (De Bruyckere et al., 2018). Here, they examine the functional implications of C1ql2 at MF synapses in Bcl11b cKO mice and following C1ql2 shRNA. The authors find that Bcl11b KO and shRNA against C1ql2 significantly reduces the recruitment of synaptic vesicles and impairs LTP at MF synapses.

Importantly, the authors test a role for the previously identified C1ql2 binding partner, exon 25b-containing Nrnx3 (Matsuda et al., 2016), as relevant at MF synapses to maintain synaptic vesicle recruitment. To test this, the authors developed a K262E C1ql2 mutant that disrupts binding to Nrnx3. Curiously, while Bcl11b KO and C1ql2 KD largely phenocopy (reduced vesicle recruitment and impaired LTP), only vesicle recruitment is dependent on C1ql2-Nrnx3 interactions. These findings provide new insight into the functional role of C1ql2 at MF synapses. While the authors convincingly demonstrate a role for C1ql2-Nrnx3(25b+) interaction for vesicle recruitment and a Nrnx3(25b+)-independent role for C1ql2 in LTP, the underlying mechanisms remain inconclusive. Additionally, a discussion of how these findings relate to previous work on C1ql2 at mossy fiber synapses and how the findings contribute to the biology of Nrnx3 would increase the interpretability of this work.

Reviewer #2 (Public Review):

This manuscript describes experiments that further investigate the actions of the transcription factor Bcl11b in regulating mossy fiber (MF) synapses in the hippocampus. Prior work from the same group had demonstrated that loss of Bcl11b results in loss of MF synapses as well as a decrease in LTP. Here the authors focus on a target of Bcl11b a secreted synaptic organizer C1ql2 which is almost completely lost in Bcl11b KO. Viral reintroduction of C1ql2 rescues the synaptic phenotypes, whereas direct KD of C1ql2 recapitulates the Bcl11b phenotype. C1ql2 itself interacts directly with Nrnx3 and replacement with a binding deficient mutant C1q was not able to rescue the Bcl11b KO phenotype. Overall there are some interesting observations in the study, however there are also some concerns about the measures and interpretation of data.

The authors state that they used a differential transcriptomic analysis to screen for candidate targets of Bcl11b, yet they do not present any details of this screen. This should be included and at the very least a table of all DE genes included. It is likely that many other genes are also regulated by Bcl11b so it would be important to the reader to see the rationale for focusing attention on C1ql2 in this study.

All viral-mediated expression uses AAVs which are known to ablate neurogenesis in the DG (Johnston DOI: 10.7554/eLife.59291) through the ITR regions and leads to hyperexcitability of the dentate. While it is not clear how this would impact the measurements the authors make in MF-CA3 synapses, this should be acknowledged as a potential caveat in this study.

The authors claim that the viral re-introduction "restored C1ql2 protein expression to control levels. This is misleading given that the mean of the data is 2.5x the control (Figure 1d and also see Figure 6c). The low n and large variance are a problem for these data. Moreover, they are marked ns but the authors should report p values for these. At the least, this likely large overexpression and variability should be acknowledged. In addition, the use of clipped bands on Western blots should be avoided. Please show the complete protein gel in primary figures of supplemental information.

Measurement of EM micrographs: As prior work suggested that MF synapse structure is disrupted the authors should report active zone length as this may itself affect "synapse score" defined by the number of vesicles docked. More concerning is that the example KO micrographs seem to have lost all the densely clustered synaptic vesicles that are away from the AZ in normal MF synapses e.g. compare control and KO terminals in Fig 2a or 6f or 7f. These terminals look aberrant and suggest that the important measure is not what is docked but what is present in the terminal cytoplasm that normally makes up the reserve pool. This needs to be addressed with further analysis and modifications to the manuscript.

The study also presents correlated changes in MF LTP in Bcl11b KO which are rescued by C1ql2 expression. It is not clear whether the structural and functional deficits are causally linked and this should be made clearer in the manuscript. It is also not apparent why this

functional measure was chosen as it is unlikely that C1ql2 plays a direct role in presynaptic plasticity mechanisms that are through a cAMP/ PKA pathway and likely disrupted LTP is due to dysfunctional synapses rather than a specific LTP effect. The authors should consider measures that might support the role of Bcl11b targets in SV recruitment during the depletion of synapses or measurements of the readily releasable pool size that would complement their findings in structural studies.

Bcl11b KO reduces the number of synapses, yet the I-O curve reported in Supp Fig 2 is not changed. How is that possible? This should be explained.

Matsuda et al DOI: 10.1016/j.neuron.2016.04.001 previously reported that C1ql2 organizes MF synapses by aligning postsynaptic kainate receptors with presynaptic elements. As this may have consequences for the functional properties of MF synapses including their plasticity, the authors should report whether they see deficient postsynaptic glutamate receptor signaling in the Bcl11b KO and rescue in the C1ql2 re-expression.

Reviewer #3 (Public Review):

Overall, this is a strong manuscript that uses multiple current techniques to provide specific mechanistic insight into prior discoveries of the contributions of the Bcl11b transcription factor to mossy fiber synapses of dentate gyrus granule cells. The authors employ an adult deletion of Bcl11b via Tamoxifen-inducible Cre and use immunohistochemical, electron microscopy, and electrophysiological studies of synaptic plasticity, together with viral rescue of C1ql2, a direct transcriptional target of Bcl11b or Nrnx3, to construct a molecular cascade downstream of Bcl11b for DG mossy fiber synapse development. They find that C1ql2 re-expression in Bcl11b cKOs can rescue the synaptic vesicle docking phenotype and the impairments in MF-LTP of these mutants. They also show that C1ql2 knockdown in DG neurons can phenocopy the vesicle docking and plasticity phenotypes of the Bcl11b cKO. They also use artificial synapse formation assays to suggest that C1ql2 functions together with a specific Nrnx3 splice isoform in mediating MF axon development, extending these data with a C1ql2-K262E mutant that purports to specifically disrupt interactions with Nrnx3. All of the molecules involved in this cascade are disease-associated and this study provides an excellent blueprint for uncovering downstream mediators of transcription factor disruption. Together this makes this work of great interest to the field. Strengths are the sophisticated use of viral replacement and multi-level phenotypic analysis while weaknesses include the linkage of C1ql2 with a specific Nrnx3 splice variant in mediating these effects.

Here is an appraisal of the main claims and conclusions:

1. C1ql2 is a downstream target of Bcl11b which mediates the synaptic vesicle recruitment and synaptic plasticity phenotypes seen in these cKOs. This is supported by the clear rescue phenotypes of synapse anatomy (Fig.2) and MF synaptic plasticity (Fig.3). One weakness here is the absence of a control assessing over-expression phenotypes of C1ql2. It's clear from Fig.1D that viral rescue is often greater than WT expression (totally expected). In the case where you are trying to suppress a LoF phenotype, it is important to make sure that enhanced expression of C1ql2 in a WT background does not cause your rescue phenotype. A strong overexpression phenotype in WT would weaken the claim that C1ql2 is the main mediator of the Bcl11b phenotype for MF synapse phenotypes.
2. Knockdown of C1ql2 via 4 shRNAs is sufficient to produce the synaptic vesicle recruitment and MF-LTP phenotypes. This is supported by clear effects in the shRNA-C1ql2 groups as compared to nonsense-EGFP controls. One concern (particularly given the use of 4 distinct shRNAs) is the potential for off-target effects, which is best controlled for by a rescue experiment with RNA-insensitive C1ql2 cDNA as opposed to nonsense sequences, which may not elicit the same off-target effects.

3. C1ql2 interacts with Nrnx3(25b+) to facilitate MF terminal SV clustering. This claim is theoretically supported by the HEK cell artificial synapse formation assay (Fig.5), the inability of the K262-C1ql2 mutation to rescue the Bcl11b phenotype (Fig.6), and the altered localization of C1ql2 in the Nrnx1-3 deletion mice (Fig.7). Each of these lines of experimental evidence has caveats that should be acknowledged and addressed. Given the hypothesis that C1ql2 and Nrnx3b(25b) are expressed in DG neurons and work together, the heterologous co-culture experiment seems strange. Up till now, the authors are looking at pre-synaptic function of C1ql2 since they are re-expressing it in DGNs. The phenotypes they are seeing are also pre-synaptic and/or consistent with pre-synaptic dysfunction. In Fig.5, they are testing whether C1ql2 can induce pre-synaptic differentiation in trans, i.e. theoretically being released from the 293 cells "post-synaptically". But the post-synaptic ligands (Nlgn1 and GluKs) are not present in the 293 cells, so a heterologous synapse assay doesn't really make sense here. The effect that the authors are seeing likely reflects the fact that C1ql2 and Nrnx3 do bind to each other, so C1ql2 is acting as an artificial post-synaptic ligand, in that it can cluster Nrnx3 which in turn clusters synaptic vesicles. But this does not test the model that the authors propose (i.e. C1ql2 and Nrnx3 are both expressed in MF terminals). Perhaps a heterologous assay where GluK2 is put into HEK cells and the C1ql2 and Nrnx3 are simultaneously or individually manipulated in DG neurons?

4. K262-C1ql2 mutation blocks the normal rescue through a Nrnx3(25b) mechanism (Fig.6). The strength of this experiment rests upon the specificity of this mutation for disrupting Nrnx3b binding (presynaptic) as opposed to any of the known postsynaptic C1ql2 ligands such as GluK2. While this is not relevant for interpreting the heterologous assay (Fig.5), it is relevant for the in vivo phenotypes in Fig.6. Similar approaches as employed in this paper can test whether binding to other known postsynaptic targets is altered by this point mutation.

5. Altered localization of C1ql2 in Nrnx1-3 cKOs. These data are presented to suggest that Nrnx3(25b) is important for localizing C1ql2 to the SL of CA3. Weaknesses of this data include both the lack of Nrnx specificity in the triple a/b KOs as well as the profound effects of Nrnx LoF on the total levels of C1ql2 protein. Some measure that isn't biased by this large difference in C1ql2 levels should be attempted (something like in Fig.1F).